复旦数学资料汇编
数理统计

来源：香蕉空间 Solution 资料库

2026 年 10 月 13 日

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第一章　数理统计

1.1　习题: 数理统计

原网页

24 春 (hyq)

参考教材:

Statistical Inference by George Casella

概率论与数理统计教程. 茆诗松

Warning: In this document, $Gamma(\alpha,\beta)$ refers to the p.d.f. $\rho(x)=\frac{1}{\beta^{\alpha}\Gamma(\alpha)}x^{\alpha-1}e^{-\frac{x}{\beta}}1_{\{x>0\}}$. This definition is same as the definition in *Statistical Inference* by George Casella, but may be different from many other books by Chinese authors. Therefore, conclusions regarding the Gamma distribution or the exponential distribution may differ by a reciprocal transformation of the second parameter β.

Chapter 2　Review of Probability and Extension

Problem 1.2

$X_1\sim Gamma(\alpha_1,\beta)$, $X_2\sim Gamma(\alpha_2,\beta)$, X_1 and X_2 are independent. Calculate the conditional distribution of $X_1|X_1+X_2=t$ for any $t>0$.

Solution: Since $X_1\sim$ Gamma (α_1,β), $X_2\sim$ Gamma (α_2,β), we know that

$$\rho_{X_1}(x_1)=\frac{1}{\beta^{\alpha_1}\Gamma(\alpha_1)}x_1^{\alpha_1-1}e^{-\frac{x_1}{\beta}}\mathbf{1}_{(0,+\infty)}(x_1),$$

$$\rho_{X_2}(x_2)=\frac{1}{\beta^{\alpha_2}\Gamma(\alpha_2)}x_2^{\alpha_2-1}e^{-\frac{x_2}{\beta}}\mathbf{1}_{(0,+\infty)}(x_2),$$

then

$$\rho_{X_1,X_2}(x_1,x_2)=\frac{1}{\beta^{\alpha_1+\alpha_2}\Gamma(\alpha_1)\Gamma(\alpha_2)}x_1^{\alpha_1-1}x_2^{\alpha_2-1}e^{-\frac{x_1+x_2}{\beta}}\mathbf{1}_{(0,+\infty)}(x_1)\mathbf{1}_{(0,+\infty)}(x_2).$$

we know that

$$\rho_{X_1,X_1+X_2}(y_1,y_2)=\rho_{X_1,X_2}(y_1,y_2-y_1)\cdot 1=\frac{y_1^{\alpha_1-1}(y_2-y_1)^{\alpha_2-1}}{\beta^{\alpha_1+\alpha_2}\Gamma(\alpha_1)\Gamma(\alpha_2)}e^{-\frac{y_2}{\alpha_2}}\mathbf{1}_{(y_2>y_1>0)}.$$

Thus

$$\rho_{X_1|X_1+X_2}(y_1\mid y_2)=\frac{\rho_{X_1,X_1+X_2}(y_1,y_2)}{\rho_{X_1+X_2}(y_2)}=\frac{\rho_{X_1,X_1+X_2}(y_1,y_2)}{\int_{\mathbb{R}}\rho_{X_1,X_1+X_2}(y_1,y_2)\,dy_1}$$

$$= \frac{y_1^{\alpha_1-1}(y_2-y_1)^{\alpha_2-1}\mathbf{1}_{\{y_2>y_1>0\}}}{\int_{\mathbb{R}} y_1^{\alpha_1-1}(y_2-y_1)^{\alpha_2-1}\mathbf{1}_{\{y_2>y_1>0\}}dy_1} = \frac{y_1^{\alpha_1-1}(y_2-y_1)^{\alpha_2-1}\mathbf{1}_{\{y_2>y_1>0\}}}{B(\alpha_1,\alpha_2)\,y_2^{\alpha_1+\alpha_2-1}}.$$

So,

$$X_1 \mid X_1+X_2=t \sim \frac{x^{\alpha_1-1}(t-x)^{\alpha_2-1}}{B(\alpha_1,\alpha_2)\,t^{\alpha_1+\alpha_2-1}}\mathbf{1}_{(0,t)}(x)\,dx.$$

Problem 1.3

For positive integer n, (use integration by parts to) show $\mathbb{P}(\text{Gamma}(n,\beta) \le x) = \mathbb{P}\left(\text{Poisson}\left(\frac{x}{\beta}\right) \ge n\right)$.

Solution:

$$\text{LHS} = \int_0^x \frac{1}{\beta^n\Gamma(n)}t^{n-1}e^{-\frac{t}{\beta}}dt = f(x)$$

$$\text{RHS} = 1-\sum_{k=0}^{n-1}\frac{\left(\frac{x}{\beta}\right)^k}{k!}e^{-\frac{x}{\beta}} = g(x) = \sum_{k=n}^{\infty}\frac{\left(\frac{x}{\beta}\right)^k}{k!}e^{-\frac{x}{\beta}}$$

$$f'(x) = \frac{1}{\beta^n\Gamma(n)}x^{n-1}e^{-\frac{x}{\beta}}$$

after some algebra.

$$g'(x) = e^{-\frac{x}{\beta}}\cdot x^{n-1}\frac{1}{\beta^n}\frac{1}{(n-1)!},$$

$$\because f'(x)=g'(x), f(0)=g(0)=0, f(x)=g(x).$$

Problem 1.5

Check (1) If $X \sim F(m,n)$, then $1/X \sim F(n,m)$; (2) If $X \sim t(n)$, then $X^2 \sim F(1,n)$; (3) If $X \sim F(m,n)$, then $\frac{m}{1+\frac{m}{n}X} \sim \text{Beta}\left(\frac{m}{2},\frac{n}{2}\right)$ (cf. Definition 1.6); (4) $\mathbb{E}\left(\frac{1}{\text{Gamma}(\alpha,\beta)}\right) = \frac{1}{(\alpha-1)\beta}$ for $\alpha>1$. Further, $\mathbb{E}\left(\frac{1}{X^2(n)}\right) = \frac{1}{n-2}$ and $\mathbb{E}(F(m,n)) = \frac{n}{n-2}$ for $n>2$.

Solution: (1)

$$X = \frac{\chi^2(m)/m}{\chi^2(n)/n} \sim F(m,n)$$

So,

$$\frac{1}{X} = \frac{\chi^2(n)/n}{\chi^2(m)/m} \sim F(n,m)$$

(2)

$$X = \frac{X_1}{\sqrt{X_2/n}} \sim t_n, \quad X_1 \sim N(0,1), X_2 \sim \chi^2(n)$$

$$X^2 = \frac{X_1^2}{X_2/n} \sim F(1,n),$$

where $X_1^2 \sim \chi^2(1)$.

(3) Let

$$X = \frac{Z/m}{Y/n}, \quad Z \sim \chi^2(m), Y \sim \chi^2(n)$$

then $\frac{m}{1+\frac{m}{n}X} = \frac{Z}{Z+Y}$. Let's consider a general transform for

$$U = Z+Y, \quad V = \frac{Z}{Z+Y}$$

where $Z \sim \text{Gamma}(\alpha_1, \beta)$ and $Y \sim \text{Gamma}(\alpha_2, \beta)$, Z and Y are independent. Note Gamma $\left(\frac{n}{2}, 2\right) = \chi^2(n)$

$$(Z, Y) \sim \frac{1}{\Gamma(\alpha_1)\Gamma(\alpha_2)\beta^{\alpha_1+\alpha_2}} z^{\alpha_1-1} y^{\alpha_2-1} e^{-\frac{z}{\beta}} e^{-\frac{y}{\beta}}, \quad z > 0, \quad y > 0.$$

Therefore,

$$(U, V) \sim \frac{1}{\Gamma(\alpha_1)\Gamma(\alpha_2)\beta^{\alpha_1+\alpha_2}} (uv)^{\alpha_1-1} [u(1-v)]^{\alpha_2-1} u e^{-\frac{u}{\beta}}$$

$$= \frac{1}{\Gamma(\alpha_1+\alpha_2)\beta^{\alpha_1+\alpha_2}} u^{\alpha_1+\alpha_2-1} e^{-\frac{u}{\beta}} \frac{\Gamma(\alpha_1+\alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} v^{\alpha_1-1}(1-v)^{\alpha_2-1}$$

Further, we have

$$U \sim \text{Gamma}(\alpha_1 + \alpha_2, \beta) \text{ and } V \sim \text{Beta}(\alpha_1, \alpha_2),$$

and U and V are independent. Particularly, $\frac{m}{1+\frac{m}{n}X} \sim$ Beta $\left(\frac{m}{2}, \frac{n}{2}\right)$.

(4)

$$E\left(\frac{1}{\text{Gamma}(\alpha, \beta)}\right) = \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^\infty \frac{1}{x} x^{\alpha-1} e^{-\frac{x}{\beta}} dx = \frac{\Gamma(\alpha-1)}{\Gamma(\alpha)} \frac{1}{\beta} = \frac{1}{\beta(\alpha-1)}$$

χ^2 distribution is Gamma $\left(\frac{n}{2}, 2\right)$, so,

$$E\left(\frac{1}{\chi^2(n)}\right) = \frac{1}{n-2}$$

$$E(F(m,n)) = E\left(\frac{Y/m}{X/n}\right) = \frac{n}{m} E(Y) E\left(\frac{1}{X}\right) = \frac{n}{m} \times m \times \frac{1}{n-2} = \frac{n}{n-2}$$

Problem 1.8

$X \mid \lambda \sim \text{Poisson}(\lambda)$ and $\lambda \sim \text{Gamma}(\alpha, \beta)$, derive the distribution of X. (It is a negative binomial if α is an integer.)

Solution: The pmf of X, for $x = 0, 1, \ldots$, is

$$f_X(x) = \int_0^\infty f_X(x \mid \lambda) f_\Lambda(\lambda) d\lambda = \int_0^\infty \frac{\lambda^x e^{-\lambda}}{x!} \frac{1}{\Gamma(\alpha)\beta^\alpha} \lambda^{\alpha-1} e^{-\lambda/\beta} d\lambda$$

$$= \frac{1}{x!\Gamma(\alpha)\beta^\alpha} \int_0^\infty \lambda^{(x+\alpha)-1} \exp\left\{-\frac{\lambda}{\left(\frac{\beta}{1+\beta}\right)}\right\} d\lambda = \frac{1}{x!\Gamma(\alpha)\beta^\alpha} \Gamma(x+\alpha) \left(\frac{\beta}{1+\beta}\right)^{x+\alpha}$$

$$= \frac{\Gamma(x+\alpha)}{x!\Gamma(\alpha)} \left(\frac{\beta}{1+\beta}\right)^x \left(\frac{1}{1+\beta}\right)^\alpha.$$

Problem 1.9

Further, $X \mid N \sim \text{Binomial}(N, p)$, $N \sim \text{Poisson}(\Lambda)$, and $\Lambda \sim \text{Gamma}(\alpha, \beta)$, find the marginal distribution of X. (three-stage model) (hint: derive $X \mid \Lambda \sim \text{Poisson}(\Lambda p)$ first)

Solution: For $x = 0, 1, \ldots$, we have

$$P(X = x \mid \lambda) = \sum_{n=x}^\infty P(X = x \mid N = n, \lambda) P(N = n \mid \lambda)$$

$$= \sum_{n=x}^{\infty} \binom{n}{x} p^x (1-p)^{n-x} \frac{e^{-\lambda}\lambda^n}{n!} = \sum_{n=x}^{\infty} \frac{1}{x!(n-x)!} \left(\frac{p}{1-p}\right)^x [(1-p)\lambda]^n e^{-\lambda}$$

$$= e^{-\lambda} \sum_{m=0}^{\infty} \frac{1}{x!m!} \left(\frac{p}{1-p}\right)^x [(1-p)\lambda]^{m+x} \quad (\text{let } m = n-x)$$

$$= \frac{1}{x!} e^{-\lambda} \left(\frac{p}{1-p}\right)^x [(1-p)\lambda]^x \sum_{m=0}^{\infty} \frac{[(1-p)\lambda]^m}{m!} = \frac{e^{-\lambda}(p\lambda)^x e^{(1-p)\lambda}}{x!} = \frac{(p\lambda)^x e^{-p\lambda}}{x!},$$

i.e. the Poisson$(p\lambda)$ pmf. Thus $X \mid \Lambda \sim$ Poisson$(p\lambda)$. Now calculations like those in Problem 1.8 yield the pmf of X, for $x = 0, 1, \ldots$, is

$$f_X(x) = \frac{1}{\Gamma(\alpha)x!(p\beta)^\alpha} \Gamma(x+\alpha) \left(\frac{p\beta}{1+p\beta}\right)^{x+\alpha}$$

$$= \frac{(\alpha+x-1)(\alpha+x-2)\cdots(\alpha+1)\alpha}{x!} \left(\frac{p\beta}{1+p\beta}\right)^x \left(\frac{1}{1+p\beta}\right)^\alpha.$$

Problem 1.10

If $X \mid \mu \sim N\left(\mu, \sigma_1^2\right)$ and $\mu \sim N\left(\delta, \sigma_2^2\right)$, check $X \sim N\left(\delta, \sigma_1^2+\sigma_2^2\right)$ and $\mu \mid X \sim N\left(\frac{\sigma_1^2}{\sigma_1^2+\sigma_2^2}\delta + \frac{\sigma_2^2}{\sigma_1^2+\sigma_2^2}X, \frac{\sigma_1^2\sigma_2^2}{\sigma_1^2+\sigma_2^2}\right)$. So the mean of posterior distribution, i.e. Bayes estimator of μ, is a weighted sum of prior mean and X.

Solution: (1)

$$E\left(e^{tX} \mid \mu\right) = e^{\mu t + \frac{1}{2}\sigma_1^2 t^2}, \quad E\left(e^{\mu t}\right) = e^{\delta t + \frac{1}{2}\sigma_2^2 t^2}$$

$$E\left(e^{tX}\right) = E\left(E\left(e^{tX} \mid \mu\right)\right) = E\left(e^{\mu t} \cdot e^{\frac{1}{2}\sigma_1^2 t^2}\right) = e^{\frac{1}{2}\sigma_1^2 t^2} \cdot E\left(e^{\mu t}\right) = e^{\delta t + \frac{1}{2}\left(\sigma_1^2+\sigma_2^2\right)t^2},$$

so $X \sim N\left(\delta, \sigma_1^2+\sigma_2^2\right)$.

(2)

$$f(\mu \mid X) = \frac{f(X,\mu)}{f(X)} = \frac{\frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(X-\mu)^2}{2\sigma_1^2}} \frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{(\mu-\delta)^2}{2\sigma_2^2}}}{\frac{1}{\sqrt{2\pi\left(\sigma_1^2+\sigma_2^2\right)}} e^{-\frac{(X-\delta)^2}{2\left(\sigma_1^2+\sigma_2^2\right)}}}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{\frac{\sigma_1^2\sigma_2^2}{\sigma_1^2+\sigma_2^2}}} \cdot \exp\left[-\frac{\left[\mu - \frac{\sigma_1^2}{\sigma_1^2+\sigma_2^2}\delta - \frac{\sigma_2^2}{\sigma_1^2+\sigma_2^2}X\right]^2}{2\frac{\sigma_1^2\sigma_2^2}{\sigma_1^2+\sigma_2^2}}\right]$$

Therefore,

$$\mu \mid X \sim N\left(\frac{\sigma_1^2}{\sigma_1^2+\sigma_2^2}\delta + \frac{\sigma_2^2}{\sigma_1^2+\sigma_2^2}X, \frac{\sigma_1^2\sigma_2^2}{\sigma_1^2+\sigma_2^2}\right).$$

Problem 1.11

Part I:

For any two r.v.s X and Y with finite variances, prove that

(a) $\text{Cov}(X, Y) = \text{Cov}(X, E(Y \mid X))$;

(b) X and $Y - E(Y \mid X)$ are uncorrelated;

(c) $\text{Var}(Y - E(Y \mid X)) = E(\text{Var}(Y \mid X))$.

Part II:

For any three r.v.s X, Y, and Z with finite variances, prove the covariance identity

$$\text{Cov}(X,Y) = E(\text{Cov}(X,Y \mid Z)) + \text{Cov}(E(X \mid Z), E(Y \mid Z)),$$

where $\text{Cov}(X,Y \mid Z)$ is the covariance of X and Y under the pdf $f(x,y \mid z)$.

Solution: (a) First, $E[XE(Y \mid X)] = E[E(XY \mid X)] = E(XY)$

$$\text{Cov}(X, E(Y \mid X)) = E[XE(Y \mid X)] - EX \cdot E[E(Y \mid X)] = E(XY) - EX \cdot EY = \text{Cov}(X,Y).$$

(b)

$$\begin{aligned}\text{Cov}(X, Y - E(Y \mid X)) &= E[XY - XE(Y \mid X)] - EX \cdot E[Y - E(Y \mid X)] \\ &= E(XY) - E(XY) - EX(EY - EY) = 0.\end{aligned}$$

(c)

$$\begin{aligned}\text{Var}(Y - E(Y \mid X)) &= E(Y - E(Y \mid X))^2 - [E(Y - E(Y \mid X))]^2 \\ &= E[Y^2 + E^2(Y \mid X) - 2YE(Y \mid X)] \\ &= EY^2 + E[E^2(Y \mid X)] - 2E[YE(Y \mid X)]\end{aligned}$$

Note that $E[YE(Y \mid X)] = E[E[YE(Y \mid X) \mid X]] = E[E(Y \mid X)E(Y \mid X)]$, so

$$\text{Var}[Y - E(Y \mid X)] = EY^2 - E[E^2(Y \mid X)].$$

As for $E(\text{Var}(Y \mid X)) = E(E(Y^2 \mid X) - E^{2(Y|X)}) = EY^2 - E[E^2(Y \mid X)]$,

we have $\text{Var}(Y - E(Y \mid X)) = E(\text{Var}(Y \mid X))$.

Part II:

$$\text{Cov}(X,Y) = E(XY) - E(X)E(Y) = E(E(XY \mid Z)) - E(E(X \mid Z))E(E(Y \mid Z))$$

$$= E(E(XY \mid Z)) - E[E(X \mid Z)E(Y \mid Z)] + E[E(X \mid Z)E(Y \mid Z)] - E(E(X \mid Z))E(E(Y \mid Z))$$

$$= E[E(XY \mid Z) - E(X \mid Z)E(Y \mid Z)] + E[E(X \mid Z)E(Y \mid Z)] - E(E(X \mid Z))E(E(Y \mid Z))$$

$$= E(\text{Cov}(X,Y \mid Z)) + \text{Cov}(E(X \mid Z), E(Y \mid Z)).$$

Problem 1.12

Assumption as in the previous "Property", prove that $ES \le \sigma$, and if $\sigma^2 > 0$, then $ES < \sigma$.

Proof. $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} (\mu, \sigma^2)$.

$$S_n^2 = \frac{1}{n-1}\left(\sum_{i=1}^n X_i^2 - n\bar{X}^2\right)$$

$$E(S_n^2) = \frac{1}{n-1}\left(\sum_{i=1}^n E(X_i^2) - nE(\bar{X}^2)\right)$$

First we calculate $E\left(\bar{X}^2\right)$.

$$E\left(\bar{X}^2\right)=\frac{1}{n^2}E\left(\sum_{i=1}^{n}X_i\right)^2=\frac{1}{n}E\left(X_1^2\right)+\frac{n-1}{n}\mu^2=\frac{\sigma^2}{n}+\mu^2.$$

So we derive that $E\left(S_n^2\right)=\sigma^2$. With Jensen inequality: given a convex function g, we have $g(EX)\leq E(g(X))$. Take $g(x)=x^2$, we can conclude that $\left(E\left(S_n\right)\right)^2\leq E\left(S_n^2\right)$, thus

$$E\left(S_n\right)\leq\sigma.$$

Further, if $\sigma>0$, then $P\left(S_n=E\left(S_n\right)\right)=0$. Note that $g(x)=x^2$ is a strictly convex function, $\left(E\left(S_n\right)\right)^2<E\left(S_n^2\right)$, thus $E\left(S_n\right)<\sigma$.

Problem 1.14

Let $X_1,X_2,\ldots,X_n$ be a random sample, and the sample mean and sample variance are calculated in the usual way.

(1) Show that $S^2=\frac{1}{2n(n-1)}\sum_{i=1}^{n}\sum_{j=1}^{n}\left(X_i-X_j\right)^2$. Assume now that the X_1 has a finite fourth moment, and denote $\theta_1=EX_1$, $\theta_j=E\left(X_1-\theta_1\right)^j$, $j=2,3,4$.

(2) Show that $\operatorname{Var}\left(S^2\right)=\frac{1}{n}\left(\theta_4-\frac{n-3}{n-1}\theta_2^2\right)$.

(3) Find $\operatorname{Cov}\left(\bar{X},S^2\right)$ in terms of $\theta_1,\theta_2,\theta_3$, and θ_4. Under what conditions is $\operatorname{Cov}\left(\bar{X},S^2\right)=0$?

Solution: (1)

$$\frac{1}{2n(n-1)}\sum_{i=1}^{n}\sum_{j=1}^{n}\left(X_i-X_j\right)^2=\frac{1}{2n(n-1)}\sum_{i=1}^{n}\sum_{j=1}^{n}\left(X_i^2+X_j^2-2X_iX_j\right)$$

$$=\frac{1}{2n(n-1)}\left[\sum_{i=1}^{n}2nX_i^2-2\sum_{i=1}^{n}X_i\sum_{j=1}^{n}X_j\right]=\frac{1}{n-1}\left(\sum_{i=1}^{n}X_i^2-n\bar{X}^2\right)$$

and

$$\begin{aligned}S^2&=\frac{1}{n-1}\sum_{i=1}^{n}(X_i-\bar{X})^2\\&=\frac{1}{n-1}\left(\sum_{i=1}^{n}X_i^2-2n\bar{X}^2+n\bar{X}^2\right)\\&=\frac{1}{n-1}\left(\sum_{i=1}^{n}X_i^2-n\bar{X}^2\right),\end{aligned}$$

so

$$S^2=\frac{1}{2n(n-1)}\sum_{i=1}^{n}\sum_{j=1}^{n}(X_i-X_j)^2.$$

(2)

$$\begin{aligned}E\left(S^2\right)&=\frac{1}{2n(n-1)}\sum_{i}\sum_{j}E(X_i-X_j)^2\\&=\frac{1}{2n(n-1)}\left(n^2-n\right)E(X_1-X_2)^2=\operatorname{Var}\left(X_1\right)=\sigma^2.\end{aligned}$$

$\text{Var}\left(S^2\right) = E\left(\left(S^2\right)^2\right) - \sigma^4$, where

$$\begin{aligned}\left(S^2\right)^2 &= \frac{1}{4n^2(n-1)^2}\left(\sum(X_i - X_j)^2\right)^2 \\ &= \frac{1}{4n^2(n-1)^2}\sum_{i\neq j}(X_i - X_j)^2\sum_{k\neq l}(X_k - X_l)^2 \\ &= \frac{1}{4n^2(n-1)^2}4\sum_{i<j}(X_i - X_j)^2\sum_{k<l}(X_k - X_l)^2 \\ &= \sum_{(i<j)\&(k<l)}\frac{1}{4n^2(n-1)^2}4(X_i - X_j)^2(X_k - X_l)^2.\end{aligned}$$

We discuss the six following situations of $(i,j),(k,l)$. Since each term is the addition and subtraction of random variables, let $\tilde{X}_i = X_i - \mu$, we assume that each X_i has been centralized, i.e. $E\left(X_i\right) = 0, \forall i, 1 \leq i \leq n$.

case (1): $(i,j) \equiv (k,l)$, $\quad E\left(X_i - X_j\right)^4 = 2\theta_4 + 6\theta_2^2$;

case (2): $k = i, j \neq l$, i.e. $(i,j)\&(i,l)$ $\quad E\left[\left(X_i - X_j\right)^2\left(X_i - X_l\right)^2\right] = \theta_4 + 3\theta_2^2$;

case (3): $l = j, i \neq k$, i.e. $(i,j)\&(k,j)$ $\quad E\left[\left(X_i - X_j\right)^2\left(X_k - X_j\right)^2\right] = \theta_4 + 3\theta_2^2$;

case (4): $i = l, j \neq k$, i.e. $(i,j)\&(k,i)$ $\quad E\left[\left(X_i - X_j\right)^2\left(X_k - X_i\right)^2\right] = \theta_4 + 3\theta_2^2$;

case (5): $j = k, i \neq l$, i.e. $(i,j)\&(j,l)$ $\quad E\left[\left(X_i - X_j\right)^2\left(X_j - X_l\right)^2\right] = \theta_4 + 3\theta_2^2$;

case (6): $i \neq k, j \neq l$, i.e. $(i,j)\&(k,l)$ $\quad E\left[\left(X_i - X_j\right)^2\left(X_k - X_l\right)^2\right] = 4\theta_2^2$.

Calculate solvability (number) of cases (1)-(6),

case (1): $\binom{n}{2}$;

case (2): $\frac{n(n-1)(n-2)}{3}$;

case (3): $\frac{n(n-1)(n-2)}{3}$;

case (4): $\frac{1}{2}\frac{n(n-1)(n-2)}{3}$;

case (5): $\frac{1}{2}\frac{n(n-1)(n-2)}{3}$;

case (6): $\binom{n}{2}\binom{n-2}{2} = \frac{n(n-1)(n-2)(n-3)}{4}$.

Then

$$\text{Var}\left(S^2\right) = \frac{1}{n}\left[\theta_4 + \frac{\theta_2^2}{n-1}\left(n^2 - 2n + 3\right) - n\theta_2^2\right] = \frac{1}{n}\left[\theta_4 - \frac{n-3}{n-1}\theta_2^2\right].$$

(3) We assume $X_i = X_i - E\left(X_i\right)$,

$$\begin{aligned}\text{Cov}\left(\bar{X}, S^2\right) &= \frac{1}{n}n\,\text{Cov}\left(X_1, S^2\right) = \text{Cov}\left(X_1, S^2\right) \\ &= \frac{1}{2n(n-1)}\left[\text{Cov}\left(X_1, (X_1 - X_2)^2\right) + \cdots + \text{Cov}\left(X_1, (X_1 - X_n)^2\right)\right] \\ &= \frac{1}{n}\text{Cov}\left(X_1, (X_1 - X_2)^2\right) = \frac{1}{n}\text{Cov}\left(X_1, X_1^2 - 2X_1X_2 + X_2^2\right) \\ &= \frac{1}{n}\left[\text{Cov}\left(X_1, X_1^2\right) - 2\,\text{Cov}\left(X_1, X_1X_2\right)\right],\end{aligned}$$

and

$$\begin{aligned}\text{Cov}\left(X_1, X_1^2\right) &= E\left(X_1^3\right) - EX_1E\left(X_1^2\right) = \theta_3, \\ \text{Cov}\left(X_1, 2X_1X_2\right) &= E\left(X_1^2X_2\right) - 0 = 0.\end{aligned}$$

So, $\text{Cov}\left(\bar{X}, S^2\right) = \frac{1}{n}\theta_3$ and if $\theta_3 = 0$, we have $\text{Cov}\left(\bar{X}, S^2\right) = 0$.

Problem 1.15 (Related to Gaussian Graphical Model)

1. Assume bivariate $\boldsymbol{X} = (X_1, X_2)^T \sim (\boldsymbol{\mu}_{2\times 1}, \boldsymbol{\Sigma}_{2\times 2})$ and let $\Sigma^{-1} = \left(\sigma^{ij}\right)_{2\times 2}$. Prove the correlation coefficient between X_1 and X_2 is $\rho_{X_1X_2} = -\sigma^{12}/\sqrt{\sigma^{11}\sigma^{22}}$.

2. Assume $\boldsymbol{X} = (X_1, X_2, \ldots, X_n)^T \sim (\boldsymbol{\mu}_{n\times 1}, \boldsymbol{\Sigma}_{n\times n})$. Find

$\max_{\beta_1^2+\beta_2^2+\ldots+\beta_n^2=1} \operatorname{Var}(\beta_1X_1 + \beta_2X_2 + \ldots + \beta_nX_n)$.

3. Assume $\boldsymbol{X} = (X_1, X_2, \ldots, X_n)^T \sim (\boldsymbol{\mu}_{n\times 1}, \boldsymbol{\Sigma}_{n\times n})$. Let $\Theta = (\theta_{ij})_{n\times n} = \Sigma^{-1}$ be the precision matrix.

Prove $-(\theta_{12}, \theta_{13}, \ldots, \theta_{1n})^\top/\theta_{11} = \operatorname{argmin}_{\beta_2,\beta_3,\ldots,\beta_n} \operatorname{Var}(X_1 - \beta_2X_2 - \beta_3X_3 - \ldots - \beta_nX_n)$.

Note $\operatorname{Var}(X_1 - \beta_2X_2 - \beta_3X_3 - \ldots - \beta_nX_n) = E(X_1 - \beta_2X_2 - \beta_3X_3 - \ldots - \beta_nX_n)^2$ when $\boldsymbol{\mu} = 0$, i.e. leading to a linear combination of $X_2, X_3, \ldots, X_n$ to approximate X_1.

4. Assume $\boldsymbol{X} = (X_1, \ldots, X_n)^T \sim N(\boldsymbol{\mu}_{n\times 1}, \boldsymbol{\Sigma}_{n\times n})$. Prove that $X_1 \perp\!\!\!\perp X_2 \mid X_3, \ldots, X_n$ (conditional independent) iff the $(1,2)$ entry of the precision matrix Θ is zero, i.e. $\theta_{1,2} = 0$.

Solution: (1)

$$\Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}, \quad \Sigma^{-1} = \frac{1}{|\Sigma|}\begin{pmatrix} \sigma_{22} & -\sigma_{12} \\ -\sigma_{21} & \sigma_{11} \end{pmatrix}$$

$$\rho_{X_1X_2} = \frac{\operatorname{Cov}(X_1, X_2)}{\sqrt{\operatorname{Var}(X_1)\operatorname{Var}(X_2)}} = \frac{\sigma_{12}}{\sqrt{\sigma_{11}\sigma_{22}}} = \frac{-|\Sigma|\sigma^{12}}{\sqrt{|\Sigma|\sigma^{22}}\sqrt{|\Sigma|\sigma^{11}}} = \frac{-\sigma^{12}}{\sqrt{\sigma^{11}\sigma^{22}}}.$$

(2)

$$\boldsymbol{\beta} = (\beta_1, \ldots, \beta_n)^\top, \quad \max_{\boldsymbol{\beta}^\top\boldsymbol{\beta}=1} \operatorname{Var}\left(\boldsymbol{\beta}^\top\boldsymbol{X}\right) = \max_{\boldsymbol{\beta}^\top\boldsymbol{\beta}=1} \boldsymbol{\beta}^\top\Sigma\boldsymbol{\beta}.$$

Let $L(\boldsymbol{\beta}, \lambda) = \boldsymbol{\beta}^\top\Sigma\boldsymbol{\beta} - \lambda\left(\boldsymbol{\beta}^\top\boldsymbol{\beta} - 1\right)$,

$$\frac{\partial L}{\partial \boldsymbol{\beta}} = 2\Sigma\boldsymbol{\beta} - 2\lambda\boldsymbol{\beta} = 0 \Rightarrow \Sigma\boldsymbol{\beta} = \lambda\boldsymbol{\beta}, \quad \boldsymbol{\beta}^\top\Sigma\boldsymbol{\beta} = \boldsymbol{\beta}^\top\lambda\boldsymbol{\beta} = \lambda.$$

Therefore, $\max_{\boldsymbol{\beta}^\top\boldsymbol{\beta}=1} \operatorname{Var}\left(\boldsymbol{\beta}^\top\boldsymbol{X}\right) = \lambda_1$ (the maximum eigenvalue of Σ).

(3) Let $\boldsymbol{\beta} = (1, -\beta_2, \ldots, -\beta_n)^\top$, then

$$\min \operatorname{Var}(X_1 - \beta_2X_2 - \ldots - \beta_nX_n) \Leftrightarrow \min \boldsymbol{\beta}^\top\Sigma\boldsymbol{\beta} \text{ s.t. } \boldsymbol{\beta}^\top \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = 1.$$

Let $L(\boldsymbol{\beta}, \lambda) = \boldsymbol{\beta}^\top\Sigma\boldsymbol{\beta} - \lambda(\boldsymbol{\beta}_1 - 1)$,

$$\begin{cases} \frac{\partial L}{\partial \boldsymbol{\beta}} = 2\Sigma\boldsymbol{\beta} - \lambda \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = 0 \Rightarrow \boldsymbol{\beta} = \frac{\lambda}{2}\Sigma^{-1} \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \frac{\lambda}{2} \begin{pmatrix} \theta_{11} \\ \theta_{12} \\ \vdots \\ \theta_{1n} \end{pmatrix} \\ \frac{\partial L}{\partial \lambda} = 0 \Rightarrow \beta_1 = 1, \lambda = \frac{2}{\theta_{11}} \end{cases}$$

Therefore, $\beta_i = -\frac{\theta_{1i}}{\theta_{11}},\ i = 2, \ldots, n$.

(4) Let $\boldsymbol{Y} = (X_3, \ldots, X_n)$, and partition

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}, \quad \Sigma_{11} = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}_{2\times 2}.$$

Then

$$(X_1, X_2) \mid \boldsymbol{Y} \sim N\left(*,\ \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} \triangleq A_{2\times 2}\right), \quad A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}.$$

$X_1 \perp\!\!\!\perp X_2 \mid \boldsymbol{Y} \Leftrightarrow a_{12} = 0 \Leftrightarrow$ the $(1,2)$ entry of A^{-1} is zero. Note that

$$\Theta = \begin{pmatrix} (\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1} & * \\ * & * \end{pmatrix},$$

so $\theta_{1,2}$ is the $(1,2)$ entry of A^{-1}. Hence $X_1 \perp\!\!\!\perp X_2 \mid \boldsymbol{Y} \Leftrightarrow \theta_{1,2} = 0$.

Chapter 3　Sample, Statistics and Distribution

Problem 2.1

Population $U[0,1]$, order statistics $U_{(1)} \le U_{(2)} \le \ldots \le U_{(n)}$. Check

(1) Joint pdf: $f(u_1, \ldots, u_n) = n!1_{\{0<u_1<u_2<\cdots<u_n<1\}}$

(2) $U_{(k)} \sim Beta(k, n-k+1)$

(3) $U_{(i+k)} - U_{(i)} \sim Beta(k, n-k+1)$ independent of i

(4) Range $U_{(n)} - U_{(1)} \sim Beta(n-1, 2)$

Solution: (1) Easy.

(2)

$$\rho_{U_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!}x^{k-1}(1-x)^{n-k}1_{[0,1]}(x).$$

So

$$\begin{aligned}\rho_{U_{(k)}}(x) &= \frac{n!}{(k-1)!(n-k)!}x^{k-1}(1-x)^{n-k}1_{[0,1]}(x) \\ &= \frac{\Gamma(n+1)}{\Gamma(k)\Gamma(n-k+1)}x^{k-1}(1-x)^{n-k+1-1}1_{[0,1]}(x) \\ &= \frac{x^{k-1}(1-x)^{n-k+1-1}}{B(k, n-k+1)}1_{[0,1]}(x).\end{aligned}$$

Hence $U_{(k)} \sim$ Beta$(k, n-k+1)$.

(3)

$$\rho_{(U_{(i)}, U_{(i+k)})}(x,y) = \frac{n!}{(i-1)!(k-1)!(n-i-k)!}x^{i-1}(y-x)^{k-1}(1-y)^{n-i-k}1_{\{0\le x<y\le 1\}}.$$

Therefore

$$\rho_{(U_{(i+k)}-U_{(i)}, U_{(i)})}(z,w) = \rho_{(U_{(i)}, U_{(i+k)})}(w, z+w)\cdot 1$$

$$= \frac{n!}{(i-1)!(k-1)!(n-i-k)!}w^{i-1}z^{k-1}(1-z-w)^{n-i-k}1_{\{w\ge 0, z\ge 0, w+z\le 1\}}.$$

Hence

$$\rho_{U_{(i+k)}-U_{(i)}}(z)=\int_{\mathbb{R}}\rho_{(U_{(i+k)}-U_{(i)},U_{(i)})}(z,w)dw$$

$$=\frac{n!}{(i-1)!(k-1)!(n-i-k)!}z^{k-1}1_{[0,1]}(z)\int_0^{1-z}w^{i-1}(1-z-w)^{n-i-k}dw$$

$$=\frac{n!}{(i-1)!(k-1)!(n-i-k)!}z^{k-1}(1-z)^{n-k}1_{[0,1]}(z)\int_0^{1}x^{i-1}(1-x)^{n-i-k}dx$$

$$=\frac{n!}{(i-1)!(k-1)!(n-i-k)!}\frac{(i-1)!(n-i-k)!}{(n-k)!}z^{k-1}(1-z)^{n-k}1_{[0,1]}(z)$$

$$=\frac{n!}{(k-1)!(n-k)!}z^{k-1}(1-z)^{n-k}1_{[0,1]}(z).$$

Namely $U_{(i+k)}-U_{(i)}\sim\text{Beta}(k,n-k+1)$.

(4) Take $i=1$ and $k=n-1$ in problem(3) and we know that $U_{(n)}-U_{(1)}\sim\text{Beta}(n-1,2)$.

Problem 2.2

Population $U_{\{1,2,\ldots,N\}}$, order statistics $U_{(1)}\le U_{(2)}\le\cdots\le U_{(n)}$. Calculate $P(U_{(1)}=k)$, $P(U_{(n)}=k)$, and $P(U_{(i)}=k)$. Further, calculate the joint pmf of $U_{(1)}$ and $U_{(n)}$.

In addition, let $U_1,U_2,\ldots,U_n$ be the sample randomly drawn from the population of $\{1:N\}$ without replacement $(n<N)$. Are the pmfs of $U_i,i=1,2,\ldots,n$ equal? Calculate $E\left[\frac{n+1}{n}U_{(n)}-1\right]$.

Solution: (1)

$$\mathbb{P}\left(U_{(1)}=k\right)=\mathbb{P}\left(U_{(1)}\ge k\right)-\mathbb{P}\left(U_{(1)}\ge k+1\right)=\left(\frac{N-k+1}{N}\right)^n-\left(\frac{N-k}{N}\right)^n.$$

$$\mathbb{P}\left(U_{(n)}=k\right)=\mathbb{P}\left(U_{(n)}\le k\right)-\mathbb{P}\left(U_{(n)}\le k-1\right)=\left(\frac{k}{N}\right)^n-\left(\frac{k-1}{N}\right)^n.$$

$$\mathbb{P}\left(U_{(i)}=k\right)=\mathbb{P}\left(U_{(i)}\le k\right)-\mathbb{P}\left(U_{(i)}\le k-1\right)$$

$$=\left[\sum_{j=i}^{n}C_n^j\left(\frac{k}{N}\right)^j\left(\frac{N-k}{N}\right)^{n-j}\right]-\left[\sum_{j=i}^{n}C_n^j\left(\frac{k-1}{N}\right)^j\left(\frac{N-k+1}{N}\right)^{n-j}\right].$$

For $1\le k\le l\le N$, if $k=l$,

$$\mathbb{P}\left(U_{(1)}=k,U_{(n)}=l\right)=\mathbb{P}\left(U_{(1)}=k,U_{(n)}=k\right)=\mathbb{P}\left(U_{(1)}=k,U_2\le k,\cdots,U_n\le k\right)$$

$$=\mathbb{P}\left(U_{(1)}=k\right)\mathbb{P}\left(U_2\le k\right)\cdots\mathbb{P}\left(U_n\le k\right)=\frac{k^{n-1}}{N^n}.$$

if $k<l$,

$$\mathbb{P}\left(U_{(1)}=k,U_{(n)}=l\right)=\mathbb{P}\left(U_{(1)}=k,\text{ the largest of }U_2\text{ to }U_n=l\right)$$

$$=\mathbb{P}\left(U_{(1)}=k\right)\mathbb{P}\left(\text{ the largest of }U_2\text{ to }U_n=l\right)$$

$$=\frac{1}{N}\left[\left(\frac{l}{N}\right)^{n-1}-\left(\frac{l-1}{N}\right)^{n-1}\right].$$

The two formulas above show the joint pmf of $U_{(1)}$ and $U_{(n)}$.

(2) Yes. And $E\left[\frac{n+1}{n}U_{(n)}-1\right]=N$.

Problem 2.3

Assume $X_1, X_2, \ldots, X_n$ is a random sample from the population of uniform distribution $U[\theta, \theta+1]$, $-\infty < \theta < \infty$. $X_{(1)}, X_{(2)}, \ldots, X_{(n)}$ are the order statistics, $R = X_{(n)} - X_{(1)}$ and $M = \left(X_{(1)} + X_{(n)}\right)/2$ are the sample range and sample midrange.

Verify:

(1) $\left(X_{(1)}, X_{(n)}\right) \sim n(n-1)\left(x_{(n)} - x_{(1)}\right)^{n-2} I_{\left\{\theta \le x_{(1)} < x_{(n)} \le \theta+1\right\}}$;

(2) $(R, M) \sim n(n-1)r^{n-2} 1_{\{0 \le r \le 1, \theta + r/2 \le m \le \theta + 1 - r/2\}}$;

(3) $R \sim n(n-1)r^{n-2}(1-r) I_{\{0 \le r \le 1\}}$, which is Beta$(n-1, 2)$. A statistic is called ancillary if its distribution doesn't involve parameter(s).

Solution: (1)

$$f(x, \theta) = I_{\{\theta \le x \le \theta+1\}},$$

$$\prod f(x_i, \theta) = \prod_{i=1}^{n} I_{\{\theta \le x_i \le \theta+1\}} = I_{\left\{X_{(1)} \ge \theta\right\}} I_{\left\{X_{(n)} \le \theta+1\right\}}.$$

$\left(X_{(1)}, X_{(n)}\right)$ are sufficient statistics,

$$f_{X_{(1)}, X_{(n)}} = n(n-1)\left(x_{(n)} - x_{(1)}\right)^{n-2} I_{\left\{\theta \le x_{(1)} < x_{(n)} \le \theta+1\right\}}.$$

(2)

$$f_{R,M}(r, m) = n(n-1)r^{n-2} I_{\left\{0 \le r \le 1, \theta + \frac{r}{2} \le m \le \theta + 1 - \frac{r}{2}\right\}}.$$

(3)

$$f_R(r) = \int f_{R,M}(r, m)dm = n(n-1)r^{n-2}(1-r) I_{\{0 \le r \le 1\}} \equiv \text{Beta}(n-1, 2),$$

which doesn't involve θ, so it is ancillary.

Problem 2.5

$X_1, ..., X_n \overset{\text{i.i.d}}{\sim} \theta + Exponential(1)$.

(1) Construct a sufficient statistic $T(X)$ for θ.

(2) If we use $T(x)$ to estimate θ, calculate the loss function $E(T(X) - \theta)^2$.

Solution: (1)

$$f(x, \theta) = \prod_{i=1}^{n} e^{-(x_i - \theta)} \mathbf{1}_{\{x_{(1)} \ge \theta\}}.$$

Then we have a sufficient statistic $X_{(1)}$ for θ.

(2)

$$E(X_{(1)} - \theta)^2 = \int_0^{+\infty} n e^{-n\sqrt{x}} \frac{1}{2\sqrt{x}} x \mathrm{d}x = \frac{2}{n^2}.$$

Problem 2.6

Find minimal sufficient statistics of Normal population $N\left(\mu,\sigma^2\right)$ with unknown μ and σ^2.

Solution: Suppose $X=(X_1,\cdots,X_n)^T\sim f(x;\theta),\theta=(\mu,\sigma^2)^T$ then

$$f\left(x_1,\cdots,x_n;\theta\right)=\prod_{k=1}^{n}\frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x_k-\mu)^2}{2\sigma^2}}=\left(\frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{\mu^2}{2\sigma^2}}\right)^n\cdot e^{-\frac{1}{2\sigma^2}\sum_{k=1}^{n}x_k^2+\frac{\mu}{\sigma^2}\sum_{k=1}^{n}x_k}.$$

Let $T(X)=\left(\sum_{k=1}^{n}x_k^2,\sum_{k=1}^{n}x_k\right)^T$, then $T(X)$ is sufficient statistic for $\left(\mu,\sigma^2\right)^T$.

Since

$$\frac{f(x;\theta)}{f(y;\theta)}=e^{-\frac{1}{2\sigma^2}\sum_{k=1}^{n}\left(x_k^2-y_k^2\right)+\frac{\mu}{\sigma^2}\sum_{k=1}^{n}(x_k-y_k)},$$

$\frac{f(x;\theta)}{f(y;\theta)}$ is independent of $\left(\mu,\sigma^2\right)^T$ iff $\sum_{k=1}^{n}\left(x_k-y_k\right)=\sum_{k=1}^{n}\left(x_k^2-y_k^2\right)=0$, which is $T(x)=T(y)$.

Thus $T(X)=\left(\sum_{k=1}^{n}x_k^2,\sum_{k=1}^{n}x_k\right)^T$ is a minimal sufficient statistic of $\theta=\left(\mu,\sigma^2\right)^T$.

$(\bar{X},S^2)$ can also be regarded as the minimal sufficient statistic of $\theta=\left(\mu,\sigma^2\right)^T$.

Problem 2.7

Find minimal sufficient statistics of

(1) Population: Exponential (λ)

(2) Population: $\theta+$ Exponential (λ) with unknown θ and λ.

Solution: (1) Suppose $X=(X_1,\cdots,X_n)^T\sim f(x;\lambda)$, then

$$f(x;\lambda)=\prod_{k=1}^{n}\lambda e^{-\lambda x_k}\mathbf{1}_{\{x_k>0\}}.$$

Let $T(X)=\sum_{k=1}^{n}X_k$, then $T(X)$ is a sufficient statistic of λ. Since

$$\frac{f(x;\lambda)}{f(y;\lambda)}=e^{-\lambda\sum_{k=1}^{n}(x_k-y_k)}\frac{1_{\{x_1>0,x_2>0,\cdots,x_n>0\}}}{1_{\{y_1>0,y_2>0,\cdots,y_n>0\}}},$$

$\frac{f(x;\lambda)}{f(y;\lambda)}$ is independent of λ iff $\sum_{k=1}^{n}(x_k-y_k)=0$, which is $T(x)=T(y)$. Thus $T(X)=\sum_{k=1}^{n}X_k$ is a minimal sufficient statistic of λ. ($\bar{X}$ is also right.)

(2) Suppose $X=(X_1,\cdots,X_n)^T\sim f\left(x;(\theta,\lambda)^T\right)$, then

$$f\left(x_1,\cdots,x_n;(\theta,\lambda)^T\right)=\prod_{k=1}^{n}\lambda e^{-\lambda(x_k-\theta)}1_{\{x_k>\theta\}}=\lambda^n e^{n\lambda\theta-\lambda\sum_{k=1}^{n}x_k}1_{\{x_1>\theta,x_2>\theta,\cdots,x_n>\theta\}}.$$

Let $T(X)=\left(\sum_{k=1}^{n}X_k,X_{(1)}\right)^T$, then $T(X)$ is a sufficient statistic of $(\theta,\lambda)^T$.

Since

$$\frac{f\left(x;(\theta,\lambda)^T\right)}{f\left(y;(\theta,\lambda)^T\right)}=e^{-\lambda\sum_{k=1}^{n}(x_k-y_k)}\frac{1_{\{x_{(1)}>\theta\}}}{1_{\{y_{(1)}>\theta\}}},$$

$\frac{f\left(x;(\theta,\lambda)^T\right)}{f\left(y;(\theta,\lambda)^T\right)}$ is independent of $(\theta,\lambda)^T$ iff $\sum_{k=1}^{n}\left(x_k-y_k\right)=x_{(1)}-y_{(1)}=0$, which is $T(x)=T(y)$.

Thus $T(X)=\left(\sum_{k=1}^{n}X_k,X_{(1)}\right)^T$ is a minimal sufficient statistic of $(\theta,\lambda)^T$.

Problem 2.9

Verify: Family of Poisson (λ), $\lambda > 0$ distribution is complete.

Solution: If $E_\lambda(g(X)) = \sum_{k=0}^{\infty} g(k)\frac{\lambda^k}{k!}e^{-\lambda} = 0$ for all $\lambda > 0$, then

$$h(\lambda) = \sum_{k=0}^{\infty} g(k)\frac{\lambda^k}{k!} = 0$$

for all $\lambda > 0$.

Let

$$f(\lambda) = \sum_{x=0}^{\infty} \frac{g(x)}{x!}\lambda^x,$$

and $f([0,\infty)) \equiv 0$, we take derivatives of $f(\lambda)$ and can get

$$\frac{g(x)}{x!} = f^{(x)}(\lambda)\frac{1}{x!} = 0 \quad \forall x \geq 0$$

thus $g(X) \equiv 0$, the distribution is complete.

Problem 2.10

Assume $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} pmf : \theta(1-\theta)^{k-1}, \theta \in (0,1), k = 1,2,3,\ldots$. Show that $\sum_{i=1}^n X_i$ is sufficient for θ, and find the family of distribution of $\sum_{i=1}^n X_i$. Is the family complete?

Solution: Suppose $X = (X_1, \cdots, X_n)^T \sim f(x;\theta)$, then

$$f(x_1, \cdots, x_n; \theta) = \prod_{k=1}^{n} \theta(1-\theta)^{x_k - 1} 1_{\{x_k \in \mathbb{Z}_+\}} = \theta^n (1-\theta)^{-n} (1-\theta)^{\sum_{k=1}^n x_k} 1_{\{x_k \in \mathbb{Z}_+, k=1,2,\cdots,n\}}.$$

So let $T(X) = \sum_{k=1}^n X_k$, then $T(X)$ is sufficient for θ.

$$f(t;\theta) = \sum_{x_1+x_2+\cdots+x_n=t} f(x_1, \cdots, x_n; \theta) = \sum_{x_1+x_2+\cdots+x_n=t} \theta^n (1-\theta)^{t-n} 1_{\{x_k \in \mathbb{Z}_+, k=1,2,\cdots,n\}} = C_{t-1}^{n-1} \theta^n (1-\theta)^{t-n}.$$

If $E_\theta(g(T)) = \sum_{t=n}^{\infty} g(t) C_{t-1}^{n-1} \theta^n (1-\theta)^{t-n}$ for all θ, then $h(\theta) = \sum_{t=n}^{\infty} g(t) C_{t-1}^{n-1} (1-\theta)^{t-n} = 0$ for all θ. Since $g(t) = \frac{1}{(t-n)! C_{t-1}^{n-1}} \lim_{\theta \to 0^+} h^{(t-n)}(\theta) = 0, t = n, n+1, \cdots$. We know the family is complete.

Problem 2.11

Assume $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} pdf : \theta x^{\theta-1}, x \in (0,1), \theta > 0$.

(1) Is $\sum_{i=1}^n X_i$ sufficient for θ ?

(2) Find a complete sufficient statistic for θ.

Solution: (1) Suppose $X = (X_1, \cdots, X_n)^T \sim f(x;\theta)$, then

$$f(x_1, \cdots, x_n; \theta) = \prod_{k=1}^{n} \theta x_k^{\theta-1} 1_{\{0<x_k<1\}} = \theta^n \left(\prod_{k=1}^{n} x_k\right)^{\theta-1} 1_{\{0<x_{(1)}<x_{(n)}<1\}}.$$

It is easy to see that $\sum_{k=1}^n X_k$ is not sufficient for θ.

(2) Let $T(X) = \prod_{k=1}^{n} X_k$, then $T(X)$ is a sufficient statistic for θ. Let $Y_k = \prod_{i=1}^{k} X_i, k = 1, 2, \cdots, n$, then

$$\begin{aligned} f_Y(y_1, y_2, \cdots, y_n; \theta) &= \frac{1}{y_1 y_2 \cdots y_{n-1}} f_X\left(y_1, \frac{y_2}{y_1}, \cdots, \frac{y_n}{y_{n-1}}; \theta\right) \\ &= \frac{1}{y_1 y_2 \cdots y_{n-1}} \theta^n y_n^{\theta-1} 1_{\{0<y_n<y_{n-1}<\cdots<y_1<1\}}. \end{aligned}$$

$$\begin{aligned} f(t;\theta) &= \int f_Y(y_1, y_2, \cdots, y_{n-1}, t; \theta)\, dy_1 dy_2 \cdots dy_{n-1} \\ &= \int \frac{1}{y_1 y_2 \cdots y_{n-1}} \theta^n t^{\theta-1} 1_{\{0<t<y_{n-1}<\cdots<y_1<1\}} dy_1 dy_2 \cdots dy_{n-1} \\ &= \theta^n t^{\theta-1} \int 1_{\{0<t, \ln t<z_{n-1}<\cdots<z_1<0\}} dz_1 dz_2 \cdots dz_{n-1} \\ &= \theta^n t^{\theta-1} \frac{(-\ln t)^{n-1}}{(n-1)!} 1_{\{0<t<1\}}. \end{aligned}$$

If

$$\begin{aligned} E_\theta(g(T)) &= \int_0^1 g(t) \theta^n t^{\theta-1} \frac{(-\ln t)^{n-1}}{(n-1)!} dt \\ &= \frac{\theta^n}{(n-1)!} \int_0^\infty g\left(e^{-s}\right) s^{n-1} e^{-s\theta} ds = 0 \end{aligned}$$

is true for every θ, with Laplace transformation, we have $g(e^{-s}) s^{n-1} = 0$ a.e., which means $T(X) = \prod_{k=1}^{n} X_k$ is complete.

Problem 2.12

Assume $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} U(\theta, 2\theta)$. Find a minimal sufficient statistic for θ. Is the statistic complete?

Solution: Suppose $X = (X_1, \cdots, X_n)^T \sim f(x;\theta)$, then

$$f(x_1, \cdots, x_n; \theta) = \prod_{k=1}^{n} \frac{1}{\theta} 1_{\{\theta<x_k<2\theta\}} = \frac{1}{\theta^n} 1_{\{\theta<x_{(1)}, x_{(n)}<2\theta\}}.$$

Let $T(X) = \left(X_{(1)}, X_{(n)}\right)^T$, then $T(X)$ is a sufficient statistic of θ. Since $\frac{f(x;\theta)}{f(y;\theta)} = \frac{1_{\{\theta<x_{(1)}, x_{(n)}<2\theta\}}}{1_{\{\theta<y_{(1)}, y_{(n)}<2\theta\}}}$, $\frac{f(x;\theta)}{f(y;\theta)}$ is independent of θ iff $x_{(1)} - y_{(1)} = x_{(n)} - y_{(n)} = 0$, which is $T(x) = T(y)$.

Thus $T(X) = \left(X_{(1)}, X_{(n)}\right)^T$ is a minimal sufficient statistic for θ.

$$f(t;\theta) = n(n-1) \left(\frac{1}{\theta}\right)^2 \left(\frac{t_2 - t_1}{\theta}\right)^{n-2} 1_{\{\theta<t_1<t_2<2\theta\}}.$$

If

$$E_\theta(g(T)) = \int g(t) f(t;\theta) dt = \int g(t) n(n-1) \left(\frac{1}{\theta}\right)^2 \left(\frac{t_2 - t_1}{\theta}\right)^{n-2} 1_{\{\theta<t_1<t_2<2\theta\}} dt_1 dt_2 = 0,$$

equivalently, $\int g(t) (t_2 - t_1)^{n-2} 1_{\{\theta<t_1<t_2<2\theta\}} dt_1 dt_2 = 0$.

Let $g(t_1, t_2) = \left(\frac{1}{t_2 - t_1}\right)^{n-2} 1_{\{t_1<t_2<kt_1\}} - \left(\frac{1}{t_2 - t_1}\right)^{n-2} 1_{\{kt_1<t_2<2t_1\}}$ where $k = \frac{9-\sqrt{17}}{4}$, then

$$\int g(t) (t_2 - t_1)^{n-2} 1_{\{\theta<t_1<t_2<2\theta\}} dt_1 dt_2 \equiv 0$$

thus T is not complete.

Problem 2.13

Assume $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \text{Gamma}(\alpha, \beta)$, where α is known, then for $T = \sum_{1\le i\le n} X_i$

$$E\left(X_{(i)} \mid T\right) = E\left[\frac{X_{(i)}}{T}T \mid T\right] = T\frac{E\left[X_{(i)}\right]}{ET}.$$

Solution: Suppose $X = (X_1, \cdots, X_n)^T \sim f(x;\beta)$, then

$$\begin{aligned} f(x_1, \cdots, x_n;\beta) &= \prod_{k=1}^{n} \frac{1}{\beta^\alpha \Gamma(\alpha)} x_k^{\alpha-1} e^{-\frac{x_k}{\beta}} \mathbf{1}_{\{x_k>0\}} \\ &= \left(\frac{1}{\beta^\alpha\Gamma(\alpha)}\right)^n \left(\prod_{k=1}^{n} x_k\right)^{\alpha-1} e^{-\frac{1}{\beta}\sum_{k=1}^n x_k} \mathbf{1}_{\{x_{(1)}>0\}}. \end{aligned}$$

Let $T(X) = \sum_{k=1}^n X_k$, then $T(X)$ is a sufficient statistic of β. By semi-generation of Gamma distribution, we have $T \sim \text{Gamma}(n\alpha, \beta)$.

If

$$E_\beta(g(T)) = \int g(t)f(t;\beta)dt = \int g(t)\frac{1}{\beta^{n\alpha}\Gamma(n\alpha)} t^{n\alpha-1}e^{-\frac{t}{\beta}}\mathbf{1}_{\{t>0\}}dt = 0, \quad \forall\beta$$

then

$$\int_0^\infty g(t)t^{n\alpha-1}e^{-\frac{t}{\beta}}dt = 0 \quad \forall\beta.$$

By Laplace transformation, we have $g(t)t^{n\alpha-1} = 0$ a.e., which means $g(t) = 0$ a.e.. Thus T is complete.

Since $\frac{X_1}{\beta}, \frac{X_2}{\beta}, \cdots, \frac{X_n}{\beta} \sim \text{Gamma}(\alpha, 1)$, $\frac{X_{(i)}}{T} = \frac{\frac{X_{(i)}}{\beta}}{\sum_{k=1}^n \frac{X_k}{\beta}}$ is ancillary statistic. By Basu's Theorem, we have T is independent of $\frac{X_{(i)}}{T}$. Thus

$$\begin{aligned} E\left(X_{(i)} \mid T\right) &= E\left(\frac{X_{(i)}}{T}T \mid T\right) = TE\left(\frac{X_{(i)}}{T} \mid T\right) \\ &= TE\left(\frac{X_{(i)}}{T}\right) = T\frac{E\left(\frac{X_{(i)}}{T}\right)ET}{ET} = T\frac{E\left(X_{(i)}\right)}{ET}. \end{aligned}$$

Chapter 4　Point and Interval Estimations

Problem 3.1

$$\begin{aligned} X &= G\gamma + g(U) + \varepsilon_X \\ Y &= X\beta + h(U) + \varepsilon_Y \\ \varepsilon_X &\perp\!\!\!\perp \varepsilon_Y; \quad \{U, \varepsilon_X, \varepsilon_Y\} \perp\!\!\!\perp G \text{ (IV: Instrumental Variable)} \end{aligned}$$

(1) Check $\beta = \text{Cov}(Y, G)/\text{Cov}(X, G)$; (2) If we have a realization of triple (G, X, Y) for n individuals, i.e. (G_i, X_i, Y_i), $i = 1, 2, \ldots, n$, what is the estimator of causal effect β ?

Solution: (1)

$$\begin{aligned} \text{Cov}(Y, G) &= E[YG] - EY \cdot EG \\ &= \beta E[XG] + EG \cdot Eh(U) + EG \cdot E\varepsilon_Y - (\beta EX + Eh(U) + E\varepsilon_Y) \cdot EG \\ &= \beta(E[XG] - EX \cdot EG) \\ &= \beta\,\text{Cov}(X, G). \end{aligned}$$

Thus $\beta = \frac{\text{Cov}(Y,G)}{\text{Cov}(X,G)}$.

(2)

$$\beta = \frac{n\sum_{i=1}^n Y_i G_i - (\sum_{i=1}^n Y_i)(\sum_{i=1}^n G_i)}{n\sum_{i=1}^n X_i G_i - (\sum_{i=1}^n X_i)(\sum_{i=1}^n G_i)}.$$

Problem 3.4: Find moment estimator

Population: (1) $U_{\{1,2,\ldots,N\}}$; (2) $\mu+$ exponential (θ), i.e. $\frac{1}{\theta}\exp\left(-\frac{x-\mu}{\theta}\right) I_{\{x>\mu\}}, \theta > 0$; (3) Binomial (m,p) with unknown m and $p \in (0,1)$.

Solution: 1. $\hat{N} = 2\bar{X} - 1$.

2.

$$EX = \int_\mu^\infty \frac{1}{\theta} e^{-\frac{x-\mu}{\theta}} x dx = \theta + \mu, \quad \text{Var}(X) = \text{Var}(\text{Exponential}(\theta)) = \theta^2.$$

Thus $\hat{\mu} = \bar{X} - \sqrt{S^2}, \hat{\theta} = \sqrt{S^2}$.

3.

$$EX = mp, \quad EX^2 = m(m-1)p^2 + mp, \quad \text{Var}(X) = EX^2 - (EX)^2 = mp(1-p).$$

Then

$$\hat{p} = 1 - \frac{S^2}{\bar{X}}, \quad \hat{m} = \frac{(\bar{X})^2}{\bar{X} - S^2}.$$

Problem 3.5: Find MLE

For population: (1) $\mu+$ exponential (θ), i.e. $\frac{1}{\theta}\exp\left(-\frac{x-\mu}{\theta}\right) I_{\{x>\mu\}}, \theta > 0$; (2) $\frac{1}{k\theta} I_{\{\theta<x<(k+1)\theta\}}$, unknown positive θ and known positive k; (3) $\frac{1}{2\theta}\exp\left(-\frac{|x|}{\theta}\right)$ with unknown positive θ; (4) $I_{\{\theta-\frac{1}{2}<x<\theta+\frac{1}{2}\}}$ (5) $\frac{1}{\theta_2-\theta_1} I_{\{\theta_1<x<\theta_2\}}$ with unknown θ_1 and θ_2.

Solution: 1.

$$L(\theta,\mu) = \frac{1}{\theta^n}\exp\left\{-\frac{1}{\theta}\left(\sum_{i=1}^n X_i - n\mu\right)\right\} \cdot \mathbf{1}_{\{X_{(1)}>\mu\}}.$$

For any fixed θ, $\hat{\mu}_{\text{MLE}} = X_{(1)}$ to insure $L(\theta,\mu)$ attains its maximum. The log-likelihood function

$$l(\theta) = \log L(\theta) = -\frac{1}{\theta}\left(\sum_{i=1}^n X_i - n\mu\right) - n\log\theta.$$

Let $\dot{l}(\theta_0) = 0$, we get $\theta_0 = \bar{X} - X_{(1)}$. Check that $\ddot{l}(\theta_0) = -\frac{n}{\left(\bar{X}-X_{(1)}\right)^2} < 0$. Thus $\hat{\theta}_{\text{MLE}} = \bar{X} - X_{(1)}, \hat{\mu}_{\text{MLE}} = X_{(1)}$.

2.

$$L(\theta) = \frac{1}{(k\theta)^n}\mathbf{1}_{\{\theta<X_{(1)}<X_{(n)}<(k+1)\theta\}}.$$

Thus $\hat{\theta}_{\text{MLE}} = \frac{X_{(n)}}{k+1}$.

3.

$$L(\theta) = \frac{1}{(2\theta)^n}\exp\left\{-\frac{1}{\theta}\sum_{i=1}^n |X_i|\right\}.$$

The log-likelihood function

$$l(\theta) = \log L(\theta) = -\frac{1}{\theta}\sum_{i=1}^n |X_i| - n\log\theta + \text{constant}.$$

Let $\dot{l}(\theta_0)=0$, we get $\theta_0=\frac{\sum_{i=1}^n|X_i|}{n}$. Check that $\ddot{l}(\theta_0)=-\frac{\sum_{i=1}^n|X_i|}{\theta_0^3}<0$. Thus $\hat{\theta}_{\text{MLE}}=\frac{\sum_{i=1}^n|X_i|}{n}$.

4.

$$L(\theta)=\mathbf{1}_{\{\theta-\frac{1}{2}<X_{(1)}<X_{(n)}<\theta+\frac{1}{2}\}}.$$

Thus $\hat{\theta}_{\text{MLE}}=\left(X_{(n)}-\frac{1}{2},X_{(1)}+\frac{1}{2}\right)$.

5.

$$L(\theta)=\frac{1}{(\theta_2-\theta_1)^n}\cdot\mathbf{1}_{\{\theta_1<X_{(1)}<X_{(n)}<\theta_2\}}.$$

Thus $\hat{\theta}_{1\text{MLE}}=X_{(1)},\hat{\theta}_{2\text{MLE}}=X_{(n)}$.

Problem 3.6

(1) Assume $X_1\sim$ Exponential (λ_1) and $X_2\sim$ Exponential (λ_2), and they are independent. Define $X=\min\{X_1,X_2\}$ and $Y=I_{\{X=X_1\}}$. Calculate the joint distribution $P\{X\le x,Y=0\}$ and $P\{X\le x,Y=1\}$. (2) For a random sample $(X_1,Y_1),(X_2,Y_2),\ldots,(X_n,Y_n)$ from (X,Y) shown in (1), calculate the MLEs of λ_1 and λ_2.

Solution: Let $Y_1=X_1-X_2,Y_2=X_2$. The joint pdf of (Y_1,Y_2) is

$$\rho\left(y_1,y_2\right)=\lambda_1\lambda_2e^{-\lambda_1y_1}e^{-(\lambda_1+\lambda_2)y_2}\cdot\mathbf{1}_{\{y_1>0,y_1+y_2>0\}}.$$

Therefore

$$\begin{aligned}P(X\le x,Y=0)&=P(Y_1>0,Y_2\le x)\\&=\lambda_1\lambda_2\int_0^x\int_0^\infty e^{-\lambda_1y_1}e^{-(\lambda_1+\lambda_2)y_2}dy_1dy_2\\&=\frac{\lambda_2}{\lambda_1+\lambda_2}\left(1-e^{-(\lambda_1+\lambda_2)x}\right)\end{aligned}$$

For the same reason, $P(X\le x,Y=1)=\frac{\lambda_1}{\lambda_1+\lambda_2}\left(1-e^{-(\lambda_1+\lambda_2)x}\right)$.

(2) Random sample $(X_i,Y_i),i=1,2,\ldots,n$. Suppose the number of $Y=1$ of samples is m, i.e. $m=\sum_{i=1}^nY_i$.

$$L(\lambda_1,\lambda_2)=\lambda_2^m\lambda_1^{n-m}\exp\left\{-\left(\lambda_1+\lambda_2\right)\sum_{i=1}^nX_i\right\}.$$

The log-likelihood function

$$l(\theta)=\log L(\theta)=m\log\lambda_2+(n-m)\log\lambda_1-(\lambda_1+\lambda_2)\sum_{i=1}^nX_i.$$

Let $\frac{\partial l}{\partial\lambda_1}=0$, $\frac{\partial l}{\partial\lambda_2}=0$. We get

$$\lambda_1=\frac{n-m}{\sum_{i=1}^nX_i},\quad\lambda_2=\frac{m}{\sum_{i=1}^nX_i}.$$

Check that $\frac{\partial^2l}{\partial\lambda_1\partial\lambda_2}=-\operatorname{diag}\left(\frac{n-m}{\lambda_1^2},\frac{m}{\lambda_2^2}\right)$ is negative. Therefore

$$\hat{\lambda}_{1\text{MLE}}=\frac{n-\sum_{i=1}^nY_i}{\sum_{i=1}^nX_i},\quad\hat{\lambda}_{2\text{MLE}}=\frac{\sum_{i=1}^nY_i}{\sum_{i=1}^nX_i}.$$

Problem 3.7

A random sample from normal population $N\left(\mu,\sigma^2\right)$. Write the likelihood function $\ell\left(\mu,\sigma^2\right)$, and for any fixed μ, find $\sigma^2(\mu)=\text{argmax}_{\sigma^2}\,\ell\left(\mu,\sigma^2\right)$ and then $\ell_p(\mu)$ and $\hat{\mu}=\text{argmax}_{\mu}\,\ell_p(\mu)$.

Solution:

$$L\left(\mu,\sigma^2\right)=\left(2\pi\sigma^2\right)^{-\frac{n}{2}}\exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i-\mu)^2\right\}.$$

The log-likelihood

$$l\left(\mu,\sigma^2\right)=\log L\left(\mu,\sigma^2\right)=-\frac{n}{2}\log 2\pi-\frac{n}{2}\log\sigma^2-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i-\mu)^2.$$

Now fix μ. Let $\frac{\partial l}{\partial\sigma^2}=0$, we get $\sigma_0^2=\frac{1}{n}\sum_{i=1}^{n}(X_i-\mu)^2$. Check that $\frac{\partial^2 l}{\partial^2\sigma^2}\left(\sigma_0^2\right)<0$, thus

$$\sigma^2(\mu)=\text{argmax}_{\sigma^2}\,l\left(\mu,\sigma^2\right)=\frac{1}{n}\sum_{i=1}^{n}(X_i-\mu)^2.$$

So

$$l_p(\mu)=l\left(\mu,\sigma^2(\mu)\right)=-\frac{n}{2}\log 2\pi-\frac{n}{2}\log\frac{1}{n}\sum_{i=1}^{n}(X_i-\mu)^2-\frac{n}{2}.$$

And

$$\hat{\mu}=\text{argmax}_{\mu}\,l_p(\mu)=\text{argmin}_{\mu}\sum_{i=1}^{n}(X_i-\mu)^2=\bar{X}.$$

Problem 3.8

Given likelihood

$$\ell(\beta,\gamma_1,\gamma_2,\ldots,\gamma_M)=\text{Constant}-\frac{1}{2}\sum_{j=1}^{M}\left[\frac{(X_j-\gamma_j)^2}{\sigma_{X_j}^2}+\frac{(Y_j-\beta\gamma_j)^2}{\sigma_{Y_j}^2}\right],$$

check

$$\ell_p(\beta)=-\frac{1}{2}\sum_{j=1}^{M}\frac{(Y_j-X_j\beta)^2}{\sigma_{X_j}^2\beta^2+\sigma_{Y_j}^2}$$

up to a constant.

Solution:

$$l(\beta,\gamma_1,\ldots,\gamma_M)=\text{Constant}-\frac{1}{2}\sum_{j=1}^{M}\left[\frac{(X_j-\gamma_j)^2}{\sigma_{X_j}^2}+\frac{(Y_j-\beta\gamma_j)^2}{\sigma_{Y_j}^2}\right].$$

Fix β, let

$$f(\gamma_1,\ldots,\gamma_M)=\sum_{j=1}^{M}\left[\frac{(X_j-\gamma_j)^2}{\sigma_{X_j}^2}+\frac{(Y_j-\beta\gamma_j)^2}{\sigma_{Y_j}^2}\right].$$

We turn to solve minimal value of f. Let

$$\frac{\partial f}{\partial\gamma_j}=\frac{2(\gamma_j-X_j)}{\sigma_{X_j}^2}+\beta\frac{2(\beta\gamma_j-Y_j)}{\sigma_{Y_j}^2}=0.$$

We attain

$$\hat{\gamma}_j = \frac{X_j\sigma_{Y_j}^2 + \beta Y_j\sigma_{X_j}^2}{\sigma_{Y_j}^2 + \beta^2\sigma_{X_j}^2}.$$

Check that $\frac{\partial^2 f}{\partial\lambda_i\partial\lambda_j} = 4\,\mathrm{diag}\left[\left(\frac{1}{\sigma_{X_j}^2} + \frac{\beta}{\sigma_{Y_j}^2}\right)_{j=1}^{n}\right]$ is positive. Therefore (up to a constant)

$$\begin{aligned}
l_p(\beta) &= -\frac{1}{2}\sum_{j=1}^{M}\left[\frac{(X_j-\gamma_j)^2}{\sigma_{X_j}^2} + \frac{(Y_j-\beta\gamma_j)^2}{\sigma_{Y_j}^2}\right] \\
&= -\frac{1}{2}\sum_{j=1}^{M}\left[\frac{\left(X_j - \frac{X_j\sigma_{Y_j}^2+\beta Y_j\sigma_{X_j}^2}{\sigma_{Y_j}^2+\beta^2\sigma_{X_j}^2}\right)^2}{\sigma_{X_j}^2} + \frac{\left(Y_j - \frac{\beta X_j\sigma_{Y_j}^2+\beta^2 Y_j\sigma_{X_j}^2}{\sigma_{Y_j}^2+\beta^2\sigma_{X_j}^2}\right)^2}{\sigma_{Y_j}^2}\right] \\
&= -\frac{1}{2}\sum_{j=1}^{M}\left[\frac{\left(\beta^2 X_j\sigma_{X_j}^2 - \beta Y_j\sigma_{X_j}\right)^2}{\sigma_{X_j}^2\left(\sigma_{Y_j}^2+\beta^2\sigma_{X_j}^2\right)^2} + \frac{\left(Y_j\sigma_{Y_j}^2 - \beta X_j\sigma_{Y_j}\right)^2}{\sigma_{Y_j}^2\left(\sigma_{Y_j}^2+\beta^2\sigma_{X_j}^2\right)^2}\right] \\
&= -\frac{1}{2}\sum_{j=1}^{M}\frac{(Y_j - X_j\beta)^2}{\left(\sigma_{Y_j}^2+\beta^2\sigma_{X_j}^2\right)^2}.
\end{aligned}$$

Problem 3.12

Assume $X_1, X_2, \ldots, X_n$ is a simple random sample from $N\left(\mu, \sigma^2\right)$, and let $A = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2, B = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2, C = \frac{1}{n+1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$. Based on MSE (mean squared error) criterion, which one is the best for estimating σ^2 ?

Solution:

$$\mathrm{MSE}(A) = \frac{2}{n-1}\sigma^4, \quad \mathrm{MSE}(B) = \frac{2n-1}{n^2}\sigma^4, \quad \mathrm{MSE}(C) = \frac{2}{n+1}\sigma^4.$$

Among them the minimal is $\mathrm{MSE}(C)$, thereby $C = \frac{1}{n+1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$ is best for estimating σ^2.

Problem 3.13

Assume $X_1, X_2, \ldots, X_n$ is a simple random sample from $U(0, \theta)$, and we already know that MLE of θ is $\hat{\theta}_{\mathrm{mle}} = X_{(n)}$ (it is also sufficient for θ), the moment estimator $2\bar{X}$ of θ, and the unbiased estimator $\frac{n+1}{n}X_{(n)}$ of θ. Which one do you prefer based on MSE? Further, find a constant c such that $\mathrm{MSE}\left[cX_{(n)}\right] < \mathrm{MSE}\left[\frac{n+1}{n}X_{(n)}\right]$ for all $\theta, n > 1$.

Solution:

$$EX_{(n)} = \frac{n}{n+1}\theta, \quad E\left[X_{(n)}^2\right] = \frac{n}{n+2}\theta^2.$$

$$\mathrm{MSE}\left(X_{(n)}\right) = \frac{2}{(n+2)(n+1)}\theta^2, \quad \mathrm{MSE}(2\bar{X}) = \frac{4}{n}\,\mathrm{Var}(X_1) = \frac{1}{3n}\theta^2,$$

$$\mathrm{MSE}\left(\frac{n+1}{n}X_{(n)}\right) = \frac{(n+1)^2}{n^2}\,\mathrm{Var}\left(X_{(n)}\right) = \frac{1}{n(n+2)}\theta^2.$$

Thus, for $n > 1$, $\frac{n+1}{n}X_{(n)}$ is better based on MSE.

$c = \frac{n+2}{n+1}$.

Problem 3.14

(1) For given n pairs of $(x_i, y_i), i = 1, \ldots, n$, let $(a, b) = \text{argmin}_{a,b} \sum_{i=1}^{n} (y_i - a - bx_i)^2$, then $b = \frac{\sum_{i=1}^{n}(x_i - \bar{x})y_i}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, a = \bar{y} - b\bar{x}$, where $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$. We call $y = a + bx$ the best line to fit $\left\{(x_i, y_i), 1 \le i \le n\right\}$. Check the validity of a and b.

(2) $Y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \epsilon_1, \epsilon_2, \ldots, \epsilon_n \overset{\text{iid}}{\sim} N\left(0, \sigma^2\right), x_1, x_2, \ldots, x_n$ are fixed constants. β_0, β_1, and σ^2 are unknown parameters of interest.

(i) Check $\left(\sum_{i=1}^{n} Y_i, \sum_{i=1}^{n} x_i Y_i, \sum_{i=1}^{n} Y_i^2\right)$ is sufficient for the unknown parameters.

(ii) Find the MLEs of unknown parameters and judge if it is unbiased.

(iii) Find the distributions of MLEs.

Solution: (1)

$$L(a, b) := \sum_{i=1}^{n} (y_i - a - bx_i)^2 .$$

Let $\frac{\partial L}{\partial a} = 0, \frac{\partial L}{\partial b} = 0$. We get

$$\hat{a} = \bar{y} - \hat{b}\bar{x}, \quad \hat{b} = \frac{\sum_{i=1}^{n} (x_i - \bar{x})\, y_i}{\sum_{i=1}^{n} (x_i - \bar{x})^2}.$$

Check that

$$\frac{\partial^2 L}{\partial a \partial b} = \begin{pmatrix} n & \sum_{i=1}^{n} x_i \\ \sum_{i=1}^{n} x_i & \sum_{i=1}^{n} x_i^2 \end{pmatrix}$$

which is nonnegative. Thus $(\hat{a}, \hat{b}) = \text{argmin}_{a,b} \sum_{i=1}^{n} (y_i - a - bx_i)^2$.

(2) $Y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \epsilon_i \sim N\left(0, \sigma^2\right)$ i.i.d..

(i) $Y_i \sim N\left(\beta_0 + \beta_1 x_i, \sigma^2\right)$ independently. Therefore the joint pdf of $y = (y_1, \ldots, y_n)$ is

$$f(y) = \left(2\pi\sigma^2\right)^{-\frac{n}{2}} \cdot \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2\right\}.$$

Considering $\sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2 = \sum_{i=1}^{n} Y_i^2 + \sum_{i=1}^{n} (\beta_0 + \beta_1 x_i)^2 - 2\sum_{i=1}^{n} Y_i (\beta_0 + \beta_1 x_i)$, by the factorization theorem, $\left(\sum_{i=1}^{n} Y_i, \sum_{i=1}^{n} x_i Y_i, \sum_{i=1}^{n} Y_i^2\right)$ is sufficient for the unknown parameters.

(ii) & (iii): Considering the matrix version of this problem we have

$$\hat{\beta} = (\hat{\beta}_1, \hat{\beta}_0)^T = (X^T X)^{-1} X^T y$$

where $X = \begin{pmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{pmatrix}_{n \times 2}$. Thus $\hat{\beta}$ is unbiased.

Let $G = X(X^T X)^{-1} X^T$, then $\hat{\sigma}^2 = \frac{y^T (I - G) y}{n} \sim \frac{\sigma^2}{n} \chi^2(n - 2)$. Thus $\hat{\sigma}^2$ is unbiased.

Problem 3.15

Population: $\theta x^{\theta - 1} I_{\{0 < x < 1\}}, \theta > 0$.

(1) Find the MLE of $\frac{1}{\theta}$;

(2) Find the efficient estimator of $\frac{1}{\theta}$.

Solution: (1) $l(\theta) = \sum_{i=1}^{n}((\theta-1)\ln X_i + \ln\theta) = n\ln\theta + (\sum_{i=1}^{n}\ln X_i)\,\theta - \sum_{i=1}^{n}\ln X_i$. Thus $\hat{\theta}_{\text{mle}} = -\frac{n}{\sum_{i=1}^{n}\ln X_i}$, and

$$\left(\frac{1}{\hat{\theta}}\right)_{\text{mle}} = \frac{1}{\hat{\theta}_{\text{mle}}} = -\frac{1}{n}\sum_{i=1}^{n}\ln X_i.$$

(2) Let $S = -\frac{1}{n}\sum_{i=1}^{n}\ln X_i \in (0,+\infty)$, then $S \sim \frac{(n\theta)^n}{(n-1)!}e^{-n\theta s}s^{n-1}ds$.

$$ES = \int_0^{+\infty} s\frac{(n\theta)^n}{(n-1)!}e^{-n\theta s}s^{n-1}\mathrm{d}s = \frac{1}{n!\theta}\int_0^{+\infty}e^{-t}t^n\mathrm{d}t = \frac{1}{\theta}.$$

Therefore S is an unbiased estimator of $\frac{1}{\theta}$.

$$ES^2 = \int_0^{+\infty} s^2\frac{(n\theta)^n}{(n-1)!}e^{-n\theta s}s^{n-1}\mathrm{d}s = \frac{1}{n\cdot n!\theta^2}\int_0^{+\infty}e^{-t}t^{n+1}\mathrm{d}t = \frac{n+1}{n\theta^2}, \text{Var}(S) = ES^2 - (ES)^2 = \frac{1}{n\theta^2}.$$

$$i\left(\frac{1}{\theta}\right) = E\Big[\left(-\theta^2\ln x - \theta\right)^2\Big] = \int_0^1 \theta x^{\theta-1}\left(\theta^2\ln x + \theta\right)^2\mathrm{d}x = \theta^2, \quad I\left(\frac{1}{\theta}\right) = ni\left(\frac{1}{\theta}\right) = n\theta^2.$$

Thus $\text{Var}(S) = \frac{1}{n\theta^2} = \frac{1}{I\left(\frac{1}{\theta}\right)}$, so $S = -\frac{1}{n}\sum_{i=1}^{n}\ln X_i$ is the efficient estimator of $\frac{1}{\theta}$.

Problem 3.16

Population: Gamma $(\alpha,\beta), \alpha > 0$ is known. Prove that $\bar{X}/\alpha$ is efficient for β.

Solution: We know

$$E[\text{Gamma}(\alpha,\beta)] = \alpha\beta, \quad \text{Var}(\text{Gamma}(\alpha,\beta)) = \alpha\beta^2.$$

The log-likelihood function

$$l(\beta) = -n\log\left(\Gamma(\alpha)\beta^\alpha\right) + (\alpha-1)\sum_{i=1}^{n}\log X_i - \frac{1}{\beta}\sum_{i=1}^{n}X_i.$$

$$\dot{l}(\beta) = -\frac{\alpha n}{\beta} + \frac{\sum_{i=1}^{n}X_i}{\beta^2}, \quad \ddot{l}(\beta) = \frac{\alpha n}{\beta^2} - \frac{2\sum_{i=1}^{n}X_i}{\beta^3}.$$

Check that

$$E\left(\frac{\bar{X}}{\alpha}\right) = \frac{1}{\alpha}E(\text{Gamma}(\alpha,\beta)) = \beta, \quad \text{Var}\left(\frac{\bar{X}}{\alpha}\right) = \frac{\text{Var}(\bar{X})}{\alpha^2} = \frac{\beta^2}{\alpha n}.$$

$$E(-\ddot{l}(\beta)) = -\frac{\alpha n}{\beta^2} + \frac{2n}{\beta^3}E(\text{Gamma}(\alpha,\beta)) = \frac{\alpha n}{\beta^2}.$$

Thus $\frac{\bar{X}}{\alpha}$ is unbiased and satisfies $\text{Var}\left(\frac{\bar{X}}{\alpha}\right) = \frac{1}{I(\beta)}$.

Problem 3.17

Population $U[0,\theta]$, Sample $X_1,\ldots,X_n$. Find MVUE of unknown parameter θ.

Solution: Population: $U[0,\theta]$. Let $T = \frac{n+1}{n}X_{(n)}$. Then

$$ET = \int_0^{\theta}\frac{n+1}{n}x\cdot n\frac{1}{\theta^n}x^{n-1}dx = \theta,$$

thus T is unbiased. Besides, for any θ, if $g(x)$ satisfies

$$\int_0^\theta g(x)\frac{n}{\theta^n}x^{n-1}dx \equiv 0.$$

Then, by taking derivatives,

$$-\int_0^\theta g(x)\frac{n^2}{\theta^{n+1}}x^{n-1}dx + \frac{n}{\theta}g(\theta) \equiv 0.$$

Thus $g(x) \equiv 0$, then $X_{(n)}$ is complete.

We also know that $X_{(n)}$ is sufficient. So by Theorem 3.6, $T = \frac{n+1}{n}X_{(n)}$ is the unique MVUE of θ.

Problem 3.18

Population: Poisson (λ)

(1) Calculate the Fisher information $i\left(\frac{1}{\lambda}\right)$;

(2) Find one function $g(\lambda)$ such that $i[g(\lambda)]$ does not dependent on λ.

Solution: Population: Poisson (λ). (1) The log-likelihood function

$$l(\lambda) = -n\frac{1}{\lambda} - \sum_{i=1}^{n} X_i \cdot \log\frac{1}{\lambda}.$$

$$\frac{\partial l}{\partial \frac{1}{\lambda}} = \lambda\left(n\lambda - \sum_{i=1}^{n} X_i\right).$$

Thus

$$I\left(\frac{1}{\lambda}\right) = E\left[\left(\frac{\partial l}{\partial \frac{1}{\lambda}}\right)^2\right] = n\lambda^3, \quad i\left(\frac{1}{\lambda}\right) = \frac{1}{n}I\left(\frac{1}{\lambda}\right) = \lambda^3.$$

(2) $g(\lambda) = \sqrt{\lambda}$ is desired.

$$l(\lambda) = -n(\sqrt{\lambda})^2 + 2\sum_{i=1}^{n} X_i \cdot \log\sqrt{\lambda}.$$

Then

$$\frac{\partial^2 l}{\partial \sqrt{\lambda}^2} = -2n - \frac{2\sum_{i=1}^{n} X_i}{\sqrt{\lambda}^2}.$$

$$i(\sqrt{\lambda}) = \frac{1}{n}E\left(\frac{\partial^2 l}{\partial \sqrt{\lambda}^2}\right) = 4.$$

Therefore, $i(g(\lambda))$ does not depend on λ.

Problem 3.19

Population: $N(\mu, 1)$. Find the MVUE of μ^2 and prove that its variance is bigger that the CRLB.

Solution:

$$l(\mu) = -\frac{1}{2}\sum_{i=1}^{n} X_i^2 + \sum_{i=1}^{n} X_i \cdot \sqrt{\mu^2} - \frac{n}{2}\mu^2 - \frac{n}{2}\log 2\pi.$$

We can derive that

$$I\left(\mu^2\right) = E\left[\frac{\partial^2 l}{(\partial\mu^2)^2}\right] = \frac{n}{4\mu^2}.$$

With factorization theorem, we know that $T = \frac{\sum_{i=1}^n X_i}{n}$ is sufficient, and $T \sim N\left(\mu, \frac{1}{n}\right)$. By Laplace transformation, we can get $N\left(\mu, \frac{1}{n}\right)$ is complete. Now let $g(x) = x^2 - \frac{1}{n}$. Then $g(T)$ is an unbiased estimator. So $g(T)$ is MVUE of μ^2. Now let $X = \sqrt{n}(T - \mu)$. Then $X^2 \sim \chi^2(1)$, thus $\text{Var}\left(X^2\right) = 2$. So

$$\text{Var}((T-\mu)^2) = \frac{2}{n}.$$

We get

$$\text{Var}(g(T)) = \text{Var}\left(T^2\right) = \frac{2}{n} + \frac{4\mu^2}{n}$$

which is strictly bigger than CRLB.

Problem 3.20

Two independent samples $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} N\left(\mu, \sigma^2\right)$ and $Y_1, Y_2, \ldots, Y_n \overset{\text{iid}}{\sim} N\left(\mu, 2\sigma^2\right)$. Find the MVUE of μ and σ^2, respectively.

Solution: For X, we have done in class. $\hat{\mu}_X = \bar{X}, \hat{\sigma^2}_X = S^2$. For $Y, \hat{\mu}_Y = \bar{Y}, \hat{\sigma}^2_Y = \frac{1}{2}S^2$.

$$f\left(y; \mu, 2\sigma^2\right) = \frac{1}{\sqrt{4\pi\sigma^2}} \exp\left(-\frac{1}{4\sigma^2}(y-\mu)^2\right),$$

$$l\left(\mu, \sigma^2\right) = -\frac{1}{2}\log 2\pi - \frac{1}{2}\log 2\sigma^2 - \frac{1}{4\sigma^2}(y-\mu)^2.$$

Then

$$E\hat{\mu}_Y = E\bar{Y} = \mu; \quad E\hat{\sigma^2}_Y = \frac{1}{2}ES^2 = \sigma^2;$$

$$\text{Var}\,\hat{\mu}_Y = \frac{1}{n}\text{Var}\,Y = \frac{2\sigma^2}{n}; \quad \text{Var}\,\hat{\sigma}^2_Y = \text{Var}\,\frac{(n-1)S^2}{2\sigma^2} = \frac{2}{n-1}\sigma^4.$$

Besides,

$$\dot{l}(\mu) = \frac{1}{2\sigma^2}(Y-\mu); \quad I(\mu) = \frac{n}{2\sigma^2};$$

$$\dot{l}(\sigma^2) = \frac{1}{2\sigma^2}[(\frac{Y-\mu}{\sqrt{2}\sigma})^2 - 1]; \quad I(\sigma^2) = \frac{n}{2\sigma^4},$$

which shows that MVUE is no smaller than CRLB.

Problem 3.21

Population: Exponential (θ). We know that the sample mean is the moment estimator, MLE, sufficient, and consistent, and also MVUE (cf. Example 3.7). Prove that there exists $\hat{\theta}$ such $\text{MSE}(\hat{\theta}) < \text{MSE}(\bar{X})$. (Hint: consider $c\bar{X}$).

Solution:

$$\begin{aligned}\text{MSE}(c\bar{X}) &= E(c\bar{X} - \theta)^2 = E\left(c^2(\bar{X})^2 - 2c\theta\bar{X} + \theta^2\right) \\ &= \frac{c^2}{n^2}E\left(X_1 + \cdots + X_n\right)^2 - \frac{2c\theta}{n}n\theta + \theta^2\end{aligned}$$

$$= \frac{c^2}{n^2}\left(2n\theta^2 + \left(n^2-n\right)\theta^2\right) - 2c\theta^2 + \theta^2$$
$$= \left(\frac{n+1}{n}c^2 - 2c + 1\right)\theta^2.$$

Let $c = 1, \frac{n}{n+1}$, then

$$\text{MSE}(\bar{X}) = \frac{\theta^2}{n}, \quad \text{MSE}(\hat{\theta}) = \frac{\theta^2}{n+1}.$$

Thus, $\hat{\theta} = \frac{n}{n+1}\bar{X}$ satisfies the requirement.

Problem 3.23

Population: Exponential(λ) = Gamma$(1,\lambda)$. Find the $1-\alpha$ CI for λ; Population: θ+Exponential(1). Find the $1-\alpha$ CI for θ.

Solution: $X \sim$ Gamma$(1,\lambda)$, $Y := \frac{X}{\lambda}$, then

$$\rho_Y(y) = \rho_X(\lambda y)\lambda = e^{-y}\cdot \mathbf{1}_{(0,\infty)}(y).$$

Thus CDF of Y is $F(y) = 1-e^{-y}$. Let y_1, y_2 solutions of $F(y_1) = \frac{\alpha}{2}, F(y_2) = 1-\frac{\alpha}{2}$, respectively. Then $y_1 = -\log\left(1-\frac{\alpha}{2}\right), y_2 = -\log\left(\frac{\alpha}{2}\right)$. We have

$$P\left(y_1 \le Y \le y_2\right) = 1-\alpha, \quad P\left(\frac{X}{y_2} \le \lambda \le \frac{X}{y_1}\right) = 1-\alpha.$$

Problem 3.24

Population: $U\left(\theta-\frac{1}{2}, \theta+\frac{1}{2}\right)$. Prove $\frac{X_{(1)}+X_{(n)}}{2} - \theta$ is a pivot and then find the $1-\alpha$ CI for θ.

Solution: Let $X \sim U\left(\theta-\frac{1}{2}, \theta+\frac{1}{2}\right), Y = X-\theta$. Then $Y \sim U\left(-\frac{1}{2}, \frac{1}{2}\right)$. Note that $\frac{X_{(1)}+X_{(n)}}{2} - \theta = \frac{Y_{(1)}+Y_{(n)}}{2}$ which is independent of θ, Thus $\frac{X_{(1)}+X_{(n)}}{2} - \theta$ is a pivot. Let $P\left(a \le \frac{Y_{(1)}+Y_{(n)}}{2} \le b\right) = 1-\alpha$, Then

$$P\left(\frac{X_{(1)}+X_{(n)}}{2} - b \le \theta \le \frac{X_{(1)}+X_{(n)}}{2} - a\right) = 1-\alpha.$$

Problem 3.25

Population: $U\left(\theta_1, \theta_2\right)$. Find the $1-\alpha$ CIs for $\theta_2-\theta_1$ and $\frac{\theta_1+\theta_2}{2}$, respectively.

Solution: Let $Y_1 = X_{(n)} - X_{(1)}, Y_2 = X_{(n)}$. We can get

$$P\left(X_{(1)} \le x_1, X_{(n)} \le x_2\right) = \frac{\left(x_2-\theta_1\right)^n - \left(x_2-x_1\right)^n}{\left(\theta_2-\theta_1\right)^n}.$$

The joint pdf of $\left(X_{(1)}, X_{(n)}\right)$ is

$$\rho_X\left(x_1, x_2\right) = \frac{n(n-1)}{\left(\theta_2-\theta_1\right)^n}\left(x_2-x_1\right)^{n-2}\mathbf{1}_{\{\theta_1 \le x_1 \le x_2 \le \theta_2\}}\left(x_1, x_2\right).$$

$$\rho_Y\left(y_1, y_2\right) = \rho_X\left(y_2-y_1, y_2\right) = \frac{n(n-1)}{\left(\theta_2-\theta_1\right)^n}y_1^{n-2}\cdot \mathbf{1}_{\{0 \le y_1, y_1+\theta_1 \le y_2 \le \theta_2\}}.$$

Thus marginal pdf of Y_1 is

$$\rho\left(y_1\right) = \int_{y_1+\theta_1}^{\theta_2} \rho_Y\left(y_1, y_2\right) dy_2 = \frac{n(n-1)}{\left(\theta_2-\theta_1\right)^{n-1}}y_1^{n-2}\left(1-\frac{y_1}{\theta_2-\theta_1}\right).$$

Then pdf of X is

$$f(x)=\frac{1}{\theta_2-\theta_1}\rho\left(\frac{x}{\theta_2-\theta_1}\right)=n(n-1)x^{n-2}(1-x)\cdot 1_{(0,1)}(x).$$

Then $\frac{X_{(n)}-X_{(1)}}{\theta_2-\theta_1}\sim \text{Beta}(n-1,2)$. The $1-\alpha$ C.I. for $\theta_2-\theta_1$ satisfies

$$1-\alpha=P\left\{L\le\theta_2-\theta_1\le U\right\}=P\left\{\frac{X_{(n)}-X_{(1)}}{U}\le\frac{X_{(n)}-X_{(1)}}{\theta_2-\theta_1}\le\frac{X_{(n)}-X_{(1)}}{L}\right\}$$

that is

$$\left(\frac{X_{(n)}-X_{(1)}}{U},\frac{X_{(n)}-X_{(1)}}{L}\right)=\left(\text{Beta}_{\frac{\alpha}{2}}(n-1,2),\text{Beta}_{1-\frac{\alpha}{2}}(n-1,2)\right),$$

$$(L,U)=\left(\frac{X_{(n)}-X_{(1)}}{\text{Beta}_{1-\frac{\alpha}{2}}(n-1,2)},\frac{X_{(n)}-X_{(1)}}{\text{Beta}_{\frac{\alpha}{2}}(n-1,2)}\right).$$

(2) Let

$$u=\frac{X_{(1)}+X_{(n)}-(\theta_1+\theta_2)}{X_{(n)}-X_{(1)}},\quad v=X_{(n)}-X_{(1)}.$$

Then the joint pdf of $\left(X_{(1)},X_{(n)}\right)$ is

$$f(x,y)=\frac{n(n-1)}{(\theta_2-\theta_1)^n}(y-x)^{n-2}1_{\{\theta_1\le x\le y\le\theta_2\}},$$

so the joint pdf of (u,v) is

$$l(u,v)=f\left(\frac{uv+\theta_1+\theta_2-v}{2},\frac{uv+v+\theta_1+\theta_2}{2}\right)\frac{|v|}{2}.$$

Thus

$$l(u,v)=\frac{n(n-1)v^{n-1}}{2(\theta_2-\theta_1)^n}1_{\left\{\theta_1\le\frac{uv+\theta_1+\theta_2-v}{2}\le\frac{uv+v+\theta_1+\theta_2}{2}\le\theta_2\right\}}.$$

Thus the pdf of U is

$$f(u)=\int l(u,v)dv=\frac{n(n-1)}{2(\theta_2-\theta_1)^n}\left(1_{\{u\ge0\}}\int_0^{\frac{\theta_1+\theta_2}{1+u}}v^{n-1}dv+1_{\{u\le0\}}\int_0^{\frac{\theta_2-\theta_1}{1-u}}v^{n-1}dv\right)=\frac{n-1}{2(1+|u|)^n}.$$

The $1-\alpha$ C.I. for $\frac{\theta_1+\theta_2}{2}$ satisfies

$$\frac{\alpha}{2}=\int_t^\infty\frac{n-1}{2(1+u)^n}du=\frac{(1+t)^{-(n-1)}}{2},$$

then $t=\alpha^{-\frac{1}{n-1}}-1$. We need

$$1-\alpha=P\left\{L\le\theta_1+\theta_2\le U\right\}=P\left\{\frac{X_{(1)}+X_{(n)}-U}{X_{(n)}-X_{(1)}}\le\frac{X_{(1)}+X_{(n)}-(\theta_1+\theta_2)}{X_{(n)}-X_{(1)}}\le\frac{X_{(1)}+X_{(n)}-L}{X_{(n)}-X_{(1)}}\right\}$$

that is

$$(L,U)=\left(X_{(1)}+X_{(n)}-\left(X_{(n)}-X_{(1)}\right)\left(\alpha^{-\frac{1}{n-1}}-1\right),X_{(1)}+X_{(n)}+\left(X_{(n)}-X_{(1)}\right)\left(\alpha^{-\frac{1}{n-1}}-1\right)\right).$$

Problem 3.26

Two independent samples $X_1, X_2, \ldots, X_m \overset{\text{iid}}{\sim} U(0, \theta_1)$ and $Y_1, Y_2, \ldots, Y_n \overset{\text{iid}}{\sim} U(0, \theta_2)$. Find the $1-\alpha$ CI for $\frac{\theta_1}{\theta_2}$. (Hint: consider $\frac{X_{(m)}}{Y_{(n)}}$)

Solution: Let $T_1 = X_{(m)}, T_2 = Y_{(n)}$, then $X = \frac{T_1}{\theta_1} \sim B(m,1), Y = \frac{T_2}{\theta_2} \sim B(n,1)$. Thus, $T = \frac{X}{Y}$ is a pivot. The pdf of T is

$$f(t) = \frac{mn}{m+n} t^{m-1}, \quad 0 < t < 1;$$
$$f(t) = \frac{mn}{m+n} t^{-n-1}, \quad t \geq 1.$$

Then the $1-\alpha$ C.I. for $\frac{\theta_1}{\theta_2}$ is

$$\left[\frac{X_{(m)}}{Y_{(n)}} \left(\frac{m+n}{m} \frac{\alpha}{2} \right)^{\frac{1}{n}}, \frac{X_{(m)}}{Y_{(n)}} \left(\frac{m+n}{n} \frac{\alpha}{2} \right)^{-\frac{1}{m}} \right].$$

Problem 3.27

For population: Bernoulli (p), we already know that $T = \sum_{i=1}^n X_i \sim \text{Binomial}(n,p)$ is sufficient for p.

(1) Use integration by parts or differentiation both sides with respect to p to prove $P\{\text{Binomial}(n,p) \geq k\} = P\{\text{Beta}(k, n-k+1) \leq p\}$.

(2) Immediate Property from definition:

$$P\{\text{Beta}(k, n-k+1) \leq p\} = P\{\text{Beta}(n-k+1, k) \geq 1-p\}, \quad F_\alpha(m,n) = \frac{1}{F_{1-\alpha}(n,m)}.$$

(3) Prove the $1-\alpha$ CI for p is

$$\left[\frac{T}{T + (n-T+1) F_{1-\frac{\alpha}{2}}(2n-2T+2, 2T)}, \frac{(T+1) F_{1-\frac{\alpha}{2}}(2T+2, 2n-2T)}{(n-T) + (T+1) F_{1-\frac{\alpha}{2}}(2T+2, 2n-2T)} \right].$$

Hint. Use property of F distribution. cf. Homework 1.5, page 27.

Solution: (1) Same as Problem 1.6.

(2) $P\{\text{Beta}(k, n-k+1) \leq p\} = P\{\text{Beta}(n-k+1, k) \geq 1-p\}$, and $F_\alpha(m,n) = \frac{1}{F_{1-\alpha}(n,m)}$.

By changing variables,

$$\begin{aligned} P\{\text{Beta}(k, n-k+1) \leq p\} &= \int_0^p \frac{1}{B(k, n-k+1)} x^{k-1} (1-x)^{n-k} dx \\ &= \int_{1-p}^1 \frac{1}{B(n-k+1, k)} (1-x)^{k-1} x^{n-k} dx \\ &= P\{\text{Beta}(n-k+1, k) \geq 1-p\}. \end{aligned}$$

Let $F \sim F(m,n)$, by definition,

$$\alpha = P\{F < F_\alpha(m,n)\} = P\left\{ \frac{1}{F} > \frac{1}{F_\alpha(m,n)} \right\}.$$

Note that $\frac{1}{F} \sim F(n,m)$, thus

$$\alpha = P\left\{ \frac{1}{F} > F_{1-\alpha}(n,m) \right\}.$$

Thus

$$F_\alpha(m,n)=\frac{1}{F_{1-\alpha}(n,m)}.$$

(3) The CDF of T is

$$G(t,p)=P_p(T\le t)=\sum_{i=1}^{[t]}C_n^ip^i(1-p)^{n-i}=\frac{\Gamma(n+1)}{\Gamma(k+1)\Gamma(n-k)}\int_p^1u^k(1-u)^{n-i}du$$

where $[t]=k$. $G(t,p)$ is continuous and strictly decreasing for p. Let $T=k$,

$$\sum_{i=0}^{k-1}C_n^ip^i(1-p)^{n-i}=1-\frac{\alpha}{2}$$

that is

$$\frac{\Gamma(n+1)}{\Gamma(k-1)\Gamma(n-k+1)}\int_0^pu^{k-1}(1-u)^{n-k}du=\frac{\alpha}{2}.$$

Note that if $B\sim\text{Beta}(m,n)$, then $F=\frac{B}{1-B}\frac{n}{m}\sim F(2m,2n)$. Here $F=\frac{p}{1-p}\frac{n-k+1}{k}\sim F(2k,2(n-k+1))$. Let

$$\frac{p}{1-p}\frac{n-k+1}{k}=F_{\alpha/2}(2k,2(n-k+1)).$$

We use T to denote k, thus

$$p_l=\frac{kF_{\alpha/2}}{kF_{\alpha/2}+(n-k+1)}=\frac{T}{T+(n-k+1)F_{1-\alpha/2}(2(n-k+1),2k)}.$$

For p_u, let

$$\sum_{i=0}^{k}C_n^ip^i(1-p)^{n-i}=\frac{\alpha}{2}.$$

Note that

$$\frac{p}{1-p}\frac{n-k}{k+1}\sim F(2(k+1),2(n-k)),$$

then $\frac{p}{1-p}\frac{n-k}{k+1}=F_{1-\alpha/2}(2(k+1),2(n-k))$. We use T to denote k, thus

$$p_u=\frac{(T+1)F_{1-\alpha/2}(2(k+1),2(n-k))}{(n-k)+(k+1)F_{1-\alpha/2}(2(k+1),2(n-k))}.$$

Chapter 5　Testing Statistical Hypothesis

Problem 4.1

Normal population: $N(\mu,\sigma^2)$.Testing for:

(1) $H_0:\mu=\mu_0$ vs $H_1:\mu\ne\mu_0$.

(2) $H_0:\sigma=\sigma_0$ vs $H_1:\sigma\ne\sigma_0$.

Problem 4.2

Two independent samples from normal populations $N(\mu_X,\sigma_X^2)$ and $N(\mu_Y,\sigma_Y^2)$, respectively.

(1) Testing for $H_0:\sigma_X^2=\sigma_Y^2$ vs $H_1:\sigma_X^2\ne\sigma_Y^2$;

(2) If $\sigma_X=\sigma_Y=\sigma$, testing for $H_0:\mu_X=\mu_Y$ vs $H_1:\mu_X\ne\mu_Y$;

(3) If $\sigma_X=\sigma_Y=\sigma$, testing for $H_0:\sigma=\sigma_0$ vs $H_1:\sigma\ne\sigma_0$.

Problem 4.7

For the simple null hypothesis $H_0 : \theta = \theta_0$, verify $p(X) \sim U(0,1)$ under H_0, where $p(x) = \inf\{\alpha \in (0,1) | \varphi_\alpha(x) = 1\}, E_{\theta_0}[\varphi_\alpha(X)] = \alpha$, φ_α is a non-randomized test function (i.e., $\varphi_\alpha(x) \in \{0,1\}$ for all x),$\{x : \varphi_{\alpha_1}(x) = 1\} \subseteq \{x : \varphi_{\alpha_2}(x) = 1\}$ when $\alpha_1 < \alpha_2$.

Solution: $\alpha = E_{\theta_0}[\varphi_\alpha(X)] = P_{\theta_0}(\varphi_\alpha(X) = 1)$

$p(x) < \alpha \Rightarrow \varphi_\alpha(x) = 1 \Rightarrow p(x) \leq \alpha$

$P_{\theta_0}(p(X) < \alpha) \leq P_{\theta_0}(\varphi_\alpha(X) = 1) = \alpha \leq P_{\theta_0}(p(X) \leq \alpha), \forall \alpha \in (0,1)$

$P(Y < \alpha) \leq \alpha \leq P(Y \leq \alpha)$

If $P(Y = \beta) = r > 0$,

$\beta + \varepsilon \geq P(Y < \beta + \varepsilon) \geq P(Y \leq \beta - \delta) + P(Y = \beta) \geq \beta - \delta + r$

Let $\varepsilon, \delta > 0, \varepsilon + \delta < r$. The contradiction was derived.

Problem 4.12

Let $G_1, \cdots, G_M$ be the genotypes of an individual at loci $1 \sim M$, respectively, and D be his/her affection status, i.e. $D = 1$ means affected and $D = 0$ means unaffected.

The logistic regression model is

$$\text{logit}P(D = 1 | G_1, \cdots, G_M) = \beta_0 + G_1\beta_1 + \cdots + G_M\beta_M.$$

Now we collect the genotypes and phenotypes for n individuals $\{(D_i, G_{i1}, \cdots, G_{iM}) | i = 1, 2, \cdots, n\}$. Derive the Score test statistic about the null hypothesis of no association between M-locus genotype and phenotype $H_0 : \beta_1 = \cdots = \beta_M = 0$.

25 春, 25 秋 (yy)

25 秋考试范围:

1. 统计量: **充分统计量-因子分解定理**, 极小充分统计量, **(关于分布族的) 完备统计量**, Basu 定理, **指数型分布族**

2. 点估计-无偏性准则: 均方误差 (MSE), **一致最小方差无偏估计 (UMVUE)**, Fisher 信息矩阵, **无偏估计的 Cramer-Rao 下界**

待估参数 $g(\theta)$ 的 **UMVUE 的 3 种求解方法**:

(a) 求出 θ 的一个完备充分统计量 T 之后, 待定可测函数 h 使得 $\mathbb{E}_\theta[h(T(X))] = g(\theta)$

(b) 求出 θ 的一个完备充分统计量 T 与 $g(\theta)$ 的一个无偏估计 $\hat{g}$ 之后, 计算条件期望 $\mathbb{E}[\hat{g}(X) \mid T(X)]$

(c) 求出 θ 的一个充分统计量 T 之后, 待定可测函数 h 使得 $\mathbb{E}_\theta[h(T(X))f(T(X))] = 0, \forall$ 可测函数 f s.t. $\mathbb{E}_\theta[f(T(X))] = 0, \mathbb{E}_\theta[(f(T(X)))^2] < +\infty$. 如果这样的 h 不存在, 则 UMVUE 不存在 (Based on Theorem 3.2 and Theorem 2.5 (ii) in *Mathematical Statistics* by Jun Shao)

3. 点估计-估计方法: **矩估计, 极大似然估计 (MLE)**

4. 假设检验-参数检验问题-最优势准则:

(1) 假设检验问题的相关基本定义:

(随机化) 检验函数 (简称“检验”in Jun Shao's book) 的势函数 (power function)、显著性水平 (level of significance)、检验水准 (size)

非随机化检验的拒绝域、检验统计量

一族以 size $\alpha \in (0,1)$ 为参数的非随机化检验函数 φ_α 相对于给定的 $x \in \mathbb{R}^n$ (这里的 n 对应到样本 X 是 n 维随机向量) 的 p-值 $p(x) := \inf\{\alpha \in (0,1) \mid \varphi_\alpha(x) = 1\}$

(2) **一致最优势检验 (UMPT)**: 简单假设检验问题的 UMPT, 单调似然比分布族的单边检验问题的 UMPT, 单参数指数型分布族的一种双边检验问题的 UMPT

(3) 一致最优势无偏检验 (UMPUT): 无偏检验、边界相似检验的定义, 多参数指数型分布族中的结论, 正态分布族中的结论

5. 假设检验-参数检验问题-检验方法:

(1) **似然比检验 (LRT)**

(2) (在零假设对应的参数空间维数小于整体参数空间维数时) **似然比统计量的渐近分布**, Wald 检验, Rao 得分检验

6. 假设检验-非参数检验问题-检验方法:

(1) 单样本中位数检验问题: Wilcoxon 符号秩检验, 两样本中位数检验问题: Mann-Whitney 检验

(2) 拟合优度检验问题:

(一维连续型分布) Kolmogorov-Smirnov 检验, Cramér-von Mises 检验, Anderson–Darling 检验

(离散型分布) Pearson χ^2 检验, 列联表独立性检验

7. 区域估计:

最精确准则: 一致最精确置信域 (UMA)

估计方法: **反转假设检验问题的拒绝域, 构造枢轴量**

—

考试范围之外但上课讲授的内容:

Wald 统计决策理论的基本概念

Bayes 估计与 Minimax 估计的基本概念

大样本理论中的一些结论 (相合性, 强相和性, 渐近分布, 即概率极限理论中的依概率, 几乎必然, 依分布收敛)

线性模型中的最小二乘估计

用 U 统计量构造无偏估计

参考教材:

[1] Mathematical Statistics by Jun Shao

其他可以参考的统计学书籍:

[2] Theory of Point Estimation by Erich Lehmann

[3] Testing Statistical Hypotheses by Erich Lehmann

[4] Mathematical Statistics: Basic Ideas and Selected Topics (Two Volumes) by Peter J. Bickel

[5] 高等统计学. 薛留根

[6] 数理统计引论. 陈希孺

25 秋期末的最后一题: 设参数 $r \in (0,+\infty)$, $\mathbb{R}^2$ 上相对于 Lebesgue 测度的密度函数族为 $f_r(x,y) = \frac{1}{\pi r^2}\mathbb{1}_{\{x^2+y^2\leqslant r^2\}}$, 取 n 个相互独立的样本 $\{(X_i,Y_i)\}_{i=1}^n \overset{\text{i.i.d}}{\sim} f_r(x,y)\,\mathrm{d}x\mathrm{d}y$.

(1) 求参数 $r \in (0,+\infty)$ 的一个完备充分统计量.

答案: $\max\limits_{1\leqslant i\leqslant n}\left(X_i^2+Y_i^2\right)$

(2) 求参数 r 的一个最小方差无偏估计.

答案: $\frac{2n+1}{2n}\sqrt{\max\limits_{1\leqslant i\leqslant n}\left(X_i^2+Y_i^2\right)}$

(3) 求参数 r 的一个置信水平为 $1-\alpha$ 的置信区间.

答案: $\left[\frac{\sqrt{\max\limits_{1\leqslant i\leqslant n}\left(X_i^2+Y_i^2\right)}}{\left(1-\frac{\alpha}{2}\right)^{\frac{1}{2n}}},\frac{\sqrt{\max\limits_{1\leqslant i\leqslant n}\left(X_i^2+Y_i^2\right)}}{\left(\frac{\alpha}{2}\right)^{\frac{1}{2n}}}\right]$

Homework 1

Due: March 16, 2025

1. Let $\begin{pmatrix}X\\Y\end{pmatrix}\sim\mathcal{N}_{k+\ell}\left(\begin{pmatrix}\mu_X\\\mu_Y\end{pmatrix},\begin{pmatrix}\Sigma_{XX}&\Sigma_{XY}\\\Sigma_{YX}&\Sigma_{YY}\end{pmatrix}\right)$, where $X\sim\mathcal{N}_k\left(\mu_X,\Sigma_{XX}\right)$.
   (a) Find the joint distribution of X and $Y-BX$, where B is a constant $\ell\times k$ matrix.
   (b) Derive the conditional distribution of Y given $X=x$.
2. Show that $\{P_\theta:\theta\in\Theta\}$ is an exponential family and find its canonical form and natural parameter space, when
   (a) P_θ is the Poisson distribution $P(\theta),\theta\in\Theta=(0,\infty)$;
   (b) P_θ is the negative binomial distribution $NB(\theta,r)$ with a fixed r, $\theta\in\Theta=(0,1)$;
   (c) P_θ is the exponential distribution $E(a,\theta)$ with a fixed $a,\theta\in\Theta=(0,\infty)$;
   (d) P_θ is the gamma distribution $\Gamma(\alpha,\gamma),\theta=(\alpha,\gamma)\in\Theta=(0,\infty)\times(0,\infty)$;
   (e) P_θ is the beta distribution $B(\alpha,\beta),\theta=(\alpha,\beta)\in\Theta=(0,1)\times(0,1)$;
   (f) P_θ is the Weibull distribution $W(\alpha,\theta)$ with a fixed $\alpha>0,\theta\in\Theta=(0,\infty)$.
3. Let $X_1,\ldots,X_n$ be i.i.d. random variables from $P_\theta\in\{P_\theta:\theta\in\Theta\}$. In the following cases, find a sufficient statistic for $\theta\in\Theta$ that has the same dimension as θ.
   (a) P_θ is the Poisson distribution $P(\theta),\theta\in(0,\infty)$.
   (b) P_θ is the negative binomial distribution $NB(\theta,r)$ with a known $r,\theta\in(0,1)$.
   (c) P_θ is the exponential distribution $E(0,\theta),\theta\in(0,\infty)$.
   (d) P_θ is the gamma distribution $\Gamma(\alpha,\gamma),\theta=(\alpha,\gamma)\in(0,\infty)\times(0,\infty)$.
   (e) P_θ is the beta distribution $B(\alpha,\beta),\theta=(\alpha,\beta)\in(0,1)\times(0,1)$.
   (f) P_θ is the log-normal distribution $LN\left(\mu,\sigma^2\right),\theta=\left(\mu,\sigma^2\right)\in\mathcal{R}\times(0,\infty)$.
   (g) P_θ is the Weibull distribution $W(\alpha,\theta)$ with a known $\alpha>0,\theta\in(0,\infty)$.
4. Let $X_1,\ldots,X_n$ be i.i.d. random variables having the exponential distribution $E(a,\theta),a\in\mathcal{R}$, and $\theta>0$. Show that the smallest order statistic, $X_{(1)}$, has the exponential distribution $E(a,\theta/n)$ and that $2\sum_{i=1}^n\left(X_i-X_{(1)}\right)/\theta$ has the chi-square distribution χ^2_{2n-2}.
5. Let $X_1,\ldots,X_n(n\geq 2)$ be i.i.d. random variables having the normal distribution $N(\theta,2)$ when $\theta=0$ and the normal distribution $N(\theta,1)$ when $\theta\in\mathcal{R}$ and $\theta\neq 0$. Show that the sample mean $\bar{X}$ is a complete statistic for θ but it is not a sufficient statistic for θ.
6. Let T and S be two statistics such that $S=\psi(T)$ for a measurable ψ. Show that
   (a) if T is complete, then S is complete;

(b) if T is complete and sufficient and ψ is one-to-one, then S is complete and sufficient;

(c) the results in (a) and (b) still hold if the completeness is replaced by the bounded completeness.

7. A discrete random variable X with

$$P(X=x)=\gamma(x)\theta^x/c(\theta),\quad x=0,1,2,\ldots$$

where $\gamma(x)\geq 0, \theta>0$, and $c(\theta)=\sum_{x=0}^{\infty}\gamma(x)\theta^x$, is called a random variable with a power series distribution.

(a) Show that $\{\gamma(x)\theta^x/c(\theta):\theta>0\}$ is an exponential family.

(b) Suppose that $X_1,\ldots,X_n$ are i.i.d. with a power series distribution $\gamma(x)\theta^x/c(\theta)$. Show that $\sum_{i=1}^n X_i$ has the power series distribution $\gamma_n(x)\theta^x/[c(\theta)]^n$, where $\gamma_n(x)$ is the coefficient of θ^x in the power series expansion of $[c(\theta)]^n$.

8. Let $Y_1,\ldots,Y_n$ be i.i.d. random variables having the Lebesgue p.d.f. $\lambda x^{\lambda-1}I_{(0,1)}(x)$ with an unknown $\lambda>0$ and let $Z_1,\ldots,Z_n$ be i.i.d. discrete random variables having the power series distribution given in Question 7 with an unknown $\theta>0$. Assume that Y_i's and Z_j's are independent. Let $X_i=Y_i+Z_i, i=1,\ldots,n$. Find a complete and sufficient statistic for the unknown parameter (θ,λ) based on the sample $X=(X_1,\ldots,X_n)$.

9. Let $X_1,\ldots,X_n$ be i.i.d. random variables having the exponential distribution $E(a,\theta)$.

(a) Show that $\sum_{i=1}^n\left(X_i-X_{(1)}\right)$ and $X_{(1)}$ are independent for any (a,θ).

(b) Show that $Z_i=\left(X_{(n)}-X_{(i)}\right)/\left(X_{(n)}-X_{(n-1)}\right), i=1,\ldots,n-2$, are independent of $\left(X_{(1)},\sum_{i=1}^n\left(X_i-X_{(1)}\right)\right)$.

Homework 2

Due: April 13, 2025

1. Let $X_1,\ldots,X_n$ be i.i.d. with Lebesgue probability density function $f_\theta(x)=\theta x^{-2}I_{(\theta,\infty)}(x)$, where $\theta>0$ is an unknown parameter. Suppose $\eta=P(X_1>t)$, where $t>0$ is a constant. Find the UMVUE of η.

2. Let $X_1,\ldots,X_n$ be i.i.d. from $N(\mu,\sigma^2)$, where $\mu\in\mathbb{R}$ is unknown and $\sigma^2>0$ is known.

(a) Find the UMVUE of μ^3 and μ^4.

(b) For a fixed $t\in\mathbb{R}$, find the UMVUE of:

$$P(X_1\leq t)\quad\text{and}\quad\frac{d}{dt}P(X_1\leq t).$$

3. Let $X_1,\ldots,X_n$ be i.i.d. uniformly distributed on the interval $(\theta_1-\theta_2,\theta_1+\theta_2)$, where $\theta_1\in\mathbb{R},\theta_2>0$, and $n>2$. Find the UMVUE of $\theta_j(j=1,2)$ and θ_1/θ_2.

4. Let $(X_1,\ldots,X_n)$ be i.i.d. with Lebesgue density

$$f_{\alpha,\beta}(x)=\alpha\beta^{-\alpha}x^{\alpha-1}I_{(0,\beta)}(x),\quad\alpha,\beta>0.$$

Find the method of moments estimator for the parameter $\theta=(\alpha,\beta)$.

5. Let $X_1,\ldots,X_n$ be i.i.d. from $N(\mu,1)$ truncated at two known points $\alpha<\beta$, i.e., the Lebesgue density of X_i is (Φ is the c.d.f. of the standard normal distribution)

$$\frac{1}{\sqrt{2\pi}[\Phi(\beta-\mu)-\Phi(\alpha-\mu)]}e^{-(x-\mu)^2/2}I_{(\alpha,\beta)}(x).$$

(a) Show that the sample mean $\bar{X}$ is an asymptotically efficient estimator for $\theta = E[X_1]$.

(b) Show that $\bar{X}$ is the unique maximum likelihood estimator (MLE) of θ.

6. Suppose n observations are obtained from the $N(\mu, 1)$ distribution (μ unknown), but only whether each observation is less than 0 is recorded. Find the MLE of μ.

7. Let $X_1, \ldots, X_n$ be i.i.d. with Lebesgue PDF $f(x; \mu, \sigma) = \frac{1}{\sigma} f\left(\frac{x-\mu}{\sigma}\right)$, where $f(x) > 0$ and $f'(x)$ exists for all $x \in \mathbb{R}, \mu \in \mathbb{R}, \sigma > 0$ (location-scale family). Denote $\theta = (\mu, \sigma)^T$. Then the Fisher information matrix contained in $X_1, \ldots, X_n$ about θ is

$$I(\theta) = \frac{n}{\sigma^2} \begin{pmatrix} \int \frac{[f'(x)]^2}{f(x)} dx & \int \frac{x f'(x) f''(x)}{f(x)} dx \\ \int \frac{x f'(x) f''(x)}{f(x)} dx & \int \frac{[x f'(x)]^2}{f(x)} dx \end{pmatrix}.$$

8. Let $X_1, \ldots, X_n$ be i.i.d. from $N(\mu, \sigma^2)$, where the mean $\mu \in \mathbb{R}$ is unknown and the variance $\sigma^2 > 0$ is known.

(a) Find the UMVUE of $\vartheta = e^{t\mu}$, where $t \neq 0$.

(b) Determine whether the variance of the UMVUE obtained in (a) attains the Cramér-Rao lower bound.

9. Let $X = (X_1, \ldots, X_n)$ be a simple sample from the Gamma distribution $\Gamma(r, \lambda)$, where r is known. Assume the prior distribution of λ is Gamma $\Gamma(b, a)$. Show that the posterior distribution of λ given X is still a Gamma distribution.

10. Draw $n = 1000$ samples $x_1, \ldots, x_n$ from the gamma distribution $\Gamma(3, 1)$. Starting from the initial value $\alpha = 1$, use the Newton-Raphson iteration method to simulate the solution of Example 8 in Lecture 5 slides.

Homework 3

Due: May 6, 2025

1. Let $X_1, \ldots, X_n$ be independent binomial random variables with success probability $p = P(X_1 = 1)$. Consider testing

$$H_0 : p = p_0 \quad \text{vs} \quad H_1 : p = p_1,$$

where $0 < p_0 < p_1 < 1$. Find the uniformly most powerful (UMP) test.

2. Let $X_1, \ldots, X_n$ be i.i.d. from a Lebesgue density f_θ, where $\theta \in \Theta \subset \mathbb{R}$. For testing $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$ at significance level α, find the UMP test when:

(a) $f_\theta(x) = \theta^{-1} e^{-x/\theta} I_{(0,\infty)}(x), \theta > 0$

(b) $f_\theta(x) = \theta^{-1} x^{\theta-1} I_{(0,1)}(x), \theta > 0$

(c) $f_\theta(x)$ is the density of $N(1, \theta)$;

(d) $f_\theta(x) = \theta^{-c} c x^{c-1} e^{-(x/\theta)^c} I_{(0,\infty)}(x), \theta > 0$, where $c > 0$ is a known constant.

3. Suppose $X_i = \beta_0 + \beta_1 t_i + \epsilon_i$, where t_i are fixed constants not all equal, ϵ_i are i.i.d. normal with mean 0 and variance σ^2, and $\beta_0, \beta_1, \sigma^2$ are unknown. Derive the uniformly most powerful unbiased (UMPU) test at level α for:

(a) $H_0 : \beta_0 \leq \theta_0$ vs $H_1 : \beta_0 > \theta_0$;

(b) $H_0 : \beta_0 = \theta_0$ vs $H_1 : \beta_0 \neq \theta_0$;

(c) $H_0 : \beta_1 \leq \theta_0$ vs $H_1 : \beta_1 > \theta_0$;

(d) $H_0 : \beta_1 = \theta_0$ vs $H_1 : \beta_1 \neq \theta_0$.

4. Let $(X_1, \ldots, X_n)$ be a random sample from $N(\mu, \sigma^2)$. Solve the following:

(a) Assume $\sigma^2 = \gamma\mu^2$ with $\gamma > 0$, and $\mu \in \mathbb{R}$ unknown. Derive the likelihood ratio test for $H_0 : \gamma = \gamma_0$ vs $H_1 : \gamma \neq \gamma_0$.

(b) By inverting the acceptance region of the LR test derived in (a), obtain a confidence interval for γ with confidence coefficient $1 - \alpha$.

5. For the following cases, construct a confidence interval with confidence coefficient $1 - \alpha_0$ using the UMP test for $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$.

(a) Let $X_1, \ldots, X_n \overset{i.i.d.}{\sim} Gamma(\nu, \theta)$, each with density $x \to \frac{1}{\Gamma(\nu)}\theta x^{\nu-1}e^{-x/\theta}I_{[x>0]}$, where $\theta > 0$ is unknown and $\nu > 0$ is known.

(b) Let $X_i \sim N(t_i, \theta, \sigma^2)$, $1 \leq i \leq n$, be independent, where t_i and $\sigma > 0$ are known.

6. Let $X = (X_1, \ldots, X_n)(n > 1)$ be a random sample from the exponential distribution on (θ, ∞) with scale parameter θ, where $\theta > 0$ is unknown; i.e., $f_\theta(x) = \theta^{-1}e^{-(x-\theta)/\theta}I_{[x>\theta]}$.

(a) Show that $\overline{X}/\theta$ and $X_{(1)}/\theta$ are both pivotal quantities, where $\overline{X}$ is the sample mean and $X_{(1)}$ is the smallest order statistic.

(b) Using the two pivotal quantities in (a), derive confidence intervals for θ with confidence coefficient $1 - \alpha$.

(c) Discuss which of the confidence intervals in (b) is superior in terms of length.

1.6　试卷: 数理统计

原网页

1.7　2026 年春季数理统计期末考试（尤翀）

注 1：可能需要的分位数表都会给出，此处不再罗列。

注 2：本卷班级平均分 70，最高分 100。

1. 设 $X_1, X_2, \ldots, X_n$ 是来自密度函数 $f(x|\theta)$ 的独立同分布样本，其中 $\theta > 0$，且

$$f(x|\theta) = \frac{1}{\theta^2}xe^{-x/\theta}, \quad x > 0$$

该分布的均值为 2θ，方差为 $2\theta^2$。

(a) 求 θ 的一个充分统计量（3 分）；

(b) 求 θ 的极大似然估计 (MLE)（4 分）；

(c) 求 θ 的矩估计 (MME)（2 分）；

(d) 求该分布单个观测的 Fisher 信息量（3 分）；

(e) MLE 的方差是否达到了 Cramér-Rao 下界？请说明理由（3 分）。

2. 设 $X_1, X_2, \ldots, X_n$ 是来自均匀分布 $U(0, \theta)$ 的独立同分布样本，其中 $\theta > 0$ 未知。令 $X_{(1)} = \min\{X_1, \ldots, X_n\}$，$X_{(n)} = \max\{X_1, \ldots, X_n\}$，并记 $Y = \frac{X_{(1)}}{X_{(n)}}$。

(a) 叙述完备统计量的定义，证明 $X_{(n)}$ 是 θ 的充分完备统计量（7 分）；

(b) 证明 Y 是辅助统计量（3 分）；

(c) 证明 $X_{(n)}$ 与 Y 相互独立（2 分）；

(d) 计算 $E\left[\left(X_{(n)}-\frac{n}{n+1}\theta\right)Y\right]$（4 分）。

3. 设 $X_1, X_2, \ldots, X_n$ 是来自泊松分布的独立同分布样本，其概率质量函数为

$$p(x|\lambda)=\frac{\lambda^x e^{-\lambda}}{x!},\quad x=0,1,2,\ldots,$$

其中 $\lambda>0$ 未知。令 $Y=\sum_{i=1}^n X_i$。

(a) 证明 Y 是 λ 的充分完备统计量（7 分）；

(b) 给出 $f(\lambda)=e^{-\lambda}$ 的一个无偏估计量，并证明（4 分）；

(c) 利用 Lehmann-Scheffé 定理以及泊松分布的可加性求 $f(\lambda)$ 的 UMVUE（4 分）。

4. 设 $X_1, X_2, \ldots, X_n$ 为来自几何分布 Geometric(p) 的独立同分布样本，其概率质量函数为

$$P(X_i=k)=(1-p)^{k-1}p,\quad k=1,2,3,\ldots,\quad 0<p<1.$$

参数 p 的先验分布为 Beta(a,b)，其密度函数为

$$\pi(p)=\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}p^{a-1}(1-p)^{b-1},\quad 0<p<1,$$

其中 $a>0, b>0$ 为已知参数。

(a) 求先验分布的期望（3 分）；

(b) 求 p 的后验分布（6 分）；

(c) 考虑后验分布的均值，求 p 的贝叶斯估计量（4 分）。

提示：Beta 函数定义为 $B(\alpha,\beta)=\int_0^1 t^{\alpha-1}(1-t)^{\beta-1}\,dt$，且 $B(\alpha,\beta)=\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$；$\Gamma(\alpha+1)=\alpha\Gamma(\alpha)$。

5. 设 $X_1, X_2, \ldots, X_n$ 是来自均匀分布 $U(0,\theta)$ 的独立同分布样本，其中 $\theta>0$ 未知，$X_{(n)}=\max\{X_1,\ldots,X_n\}$。

(a) 计算 $Y=\frac{X_{(n)}}{\theta}$ 的分布（4 分）；

(b) 利用 (1) 中的枢轴量，求 θ 的置信水平为 $1-\alpha$ 的等尾置信区间（7 分）。

6. 设随机变量 X 的分布密度可取自下面的 $f_0(x)$ 或 $f_1(x)$：

$$f_0(x)=\begin{cases}1, & 0\le x\le 1,\\ 0, & \text{其他},\end{cases}$$

$$f_1(x)=\begin{cases}\frac{C}{1+x}, & 0\le x\le 1,\\ 0, & \text{其他},\end{cases}$$

其中 C 为未知常数。基于 X 的一个观测值，考虑检验问题

$$H_0: f(x)=f_0(x)\quad \text{vs}\quad H_1: f(x)=f_1(x).$$

(a) 计算 C 的值（3 分）；

(b) 利用 Neyman-Pearson 引理，求上述检验问题的检验水平 $\alpha=0.05$ 的最大功效检验 ϕ 的拒绝域（5 分）；

(c) 求 ϕ 的第二类错误的概率（5 分）。

7. 某饮料商用两种不同配方推出了两种新的饮料，现抽取了 10 位消费者，让他们分别品尝两种饮料并加以评分，从不喜欢到喜欢，评分为 1 到 10 的整数，评分结果如下：

品尝者	1	2	3	4	5	6	7	8	9	10
A 饮料	10	8	6	8	7	5	1	3	10	7
B 饮料	6	5	2	2	4	6	4	5	9	8
差值	4	3	4	6	3	-1	-3	-2	1	-1

显著性水平 $\alpha = 0.05$，问两种饮料评分是否有显著差异？

(a) 假设评分服从正态分布，对差值的均值进行检验（5 分）；

(b) 采用符号检验方法作检验（6 分）；

(c) 采用符号秩和检验方法作检验（6 分）。

1.8　2026 年春季数理统计期末考试（杨莹）

1. 设 X_1, X_2 是来自几何分布（参数为 p）的独立样本，即

$$P(X_i = x) = (1-p)^{x-1}p, \quad x = 1, 2, \ldots$$

考虑假设检验：

$$H_0 : p = 0.5, \qquad H_1 : p = 0.25$$

拒绝域取为

$$W = \{(x_1, x_2) : x_1 + x_2 \geq c\}$$

(a) 求势函数 $\beta(p)$；

(b) 若显著性水平 $\alpha = 0.1$，求临界值 c；

(c) 取第 (2) 问中的 c，计算犯第二类错误的概率。

2. 设 $X_1, \ldots, X_n$ 是来自双指数分布（拉普拉斯分布）的独立样本，密度为

$$f(x;\theta) = \frac{1}{2\theta}\exp\left(-\frac{|x|}{\theta}\right), \quad \theta > 0$$

(a) 求 θ 的 MLE；

(b) 求 UMVUE 并论证是否达到 Cramér-Rao 下界；

(c) 假设检验 $H_0 : \theta = \theta_0$，求关于大样本似然比统计量的置信区间。

3. 考虑指数族分布

$$f(x;\theta)=\exp\{\theta x-b(\theta)\}h(x)$$

其中 $b(x)>0$ 是凸函数且二阶可导。

(a) 论证这是单调似然比（MLR）分布族；

(b) 判断 UMP 检验是否存在；

(c) 给出 UMP 检验的具体形式；

(d) 用 Poisson 分布验证第 (3) 问的结果。

4. 给定 $X_1,\dots,X_{10}$，$Y_1,\dots,Y_{12}$，分别来自两个正态分布，样本均值为 $\bar{X}=73.4$，$\bar{Y}=67.8$，样本方差为 $S_1^2=36$，$S_2^2=25$。

(a) 进行方差相等假设检验；

(b) 判断能否说明两总体均值不同。

（需给出假设、统计量、拒绝域和计算结果）

5. 设 $X_1,\dots,X_n$ 独立同分布，密度为

$$f(x;\theta)=\frac{1}{\theta}\exp\left(-\frac{x-\theta}{\theta}\right),\quad x>\theta,\ \theta>0$$

(a) 证明 $\bar{X}/\theta$ 和 $X_{(1)}/\theta$ 都是枢轴量；

(b) 分别给出 θ 的 α 置信区间；

(c) 利用区间长度说明哪一个较好。

6. 给定一个 3×2 列联表，利用卡方检验判断行变量与列变量是否有关（$\alpha=0.05$）。（具体数据已经忘了) (需给出假设、统计量、拒绝域和计算结果)

1.9　2026 年春季数理统计期中考试（尤翀）

1. 某电影院拥有 N 个座位, 与另一家影院竞争 2400 名潜在观众. 假设每名观众选择该影院的概率为 0.4, 且各人的选择相互独立.

(1) 利用二项分布建模, 写出 “因影院客满导致观众无法入场” 的概率表达式 (无需计算具体数值).

(2) 利用正态近似求出满足该概率小于 2% 的最小 N 值. (可能用到的结论: $z_{0.02}=2.05$, 满足 $P(Z>z_{0.02})=0.02$, 其中 $Z\sim N(0,1)$)

2. 设离散随机变量 X 取值为非负整数, 定义其生成函数 $G_X(s)=\mathbb{E}[s^X]$, 令 $K(s)=\log(G_X(s))$.

(1) 证明: $\left.\dfrac{d}{ds}K(s)\right|_{s=1}=\mathbb{E}[X]$;

(2) 求 $\left.\dfrac{d^2}{ds^2}K(s)\right|_{s=1}$ 的表达式 (用 $\mathbb{E}[X]$ 和 $\mathrm{Var}(X)$ 表示).

3. 设随机变量 Y 表示 n 次独立伯努利试验中成功的次数, 每次试验的成功概率为 X, 其中 $n\geq 1$ 为已知常数. 假设 X 是一个连续随机变量, 其概率密度函数为

$$f_X(x)=2x,\quad 0<x<1.$$

(1) 推导 $\mathrm{Var}(Y)=\mathrm{Var}(\mathbb{E}[Y|X])+\mathbb{E}(\mathrm{Var}[Y|X])$.

(2) 求 $\mathbb{E}[Y]$ 和 $\mathrm{Var}(Y)$.

(3) 写出 X 与 Y 的联合概率密度 / 分布列 $f_{X,Y}(x,y)$.

(4) 求 Y 的边缘分布列 $P(Y=y), y=0,1,\ldots,n$. 提示: Beta 函数 $B(\alpha,\beta)=\int_0^1 x^{\alpha-1}(1-x)^{\beta-1}\,dx$, 且当 α,β 为正整数时, $B(\alpha,\beta)=\dfrac{(\alpha-1)!(\beta-1)!}{(\alpha+\beta-1)!}$.

4. 设随机样本 $X_1,X_2,\ldots,X_n$ 独立同分布, 服从参数为 $\theta>0$ 的指数分布, 其概率密度函数为

$$f_X(x)=\frac{1}{\theta}e^{-x/\theta},\quad x>0.$$

记 $X_{(1)}=\min\{X_1,\ldots,X_n\}$ 为最小次序统计量, 并令 $\bar{X}=\dfrac{1}{n}\sum_{i=1}^n X_i$ 为样本均值. 证明: $X_{(1)}/\bar{X}$ 与 $\bar{X}$ 相互独立.

提示 1: $Z_i/\theta\sim\exp(1)$

提示 2: $S=\sum_{i=1}^n X_i\sim\text{Gamma}(n,\theta)$, 对应的 pdf 为 $f(s;n,\theta)=\dfrac{1}{\Gamma(n)\theta^n}s^{n-1}e^{-s/\theta}$

提示 3: 若一个函数 h 的拉普拉斯变换在某个区间上存在且恒为零, 则 $h(s)=0$ 几乎处处 (关于 Lebesgue 测度).

1.10　2026 年春季数理统计期中考试（杨莹）

评分表：	1	2	3	4	5
	8+7	10+10	10+10+10	15	10+10

1. 设 $X_1,\ldots,X_n$ i.i.d 服从两点分布 $b(1,p)$，$0<p<1$ 是未知参数，求：
(a) p^s 的 UMVUE；
(b) $p^s+(1-p)^{n-s}$ 的 UMVUE（$0<s<n$ 是整数）。

2. 假设 $X_1,\ldots,X_n$ 是独立同分布的，服从参数为 λ 的泊松分布，即 $P(X=n)=\frac{e^{-\lambda}\lambda^n}{n!}$。
(a) 证明：存在 $g(\lambda)$ 的无偏估计量的充分必要条件是 $g(\lambda)$ 是一个收敛级数 $\sum_{k\geq 0}a_k\lambda^k$，$0<\lambda<\infty$。
(b) 令 $T=\sum_{k=1}^n X_k$，找到一个关于 T 的函数，作为 $e^{-2\lambda}$ 的无偏估计量，并证明它也是 $e^{-2\lambda}$ 的 UMVUE。当 $n=1$ 时，这是一个好的估计量吗？

3. 设 $X_1,\ldots,X_n$ 是独立同分布的样本，对于某个 $0<\theta<1$，有 $P(X_1=-1)=\frac{1-\theta}{2}, P(X_1=0)=\frac{1}{2}, P(X_1=1)=\frac{\theta}{2}$。
(a) 推导 θ 的最大似然估计量 $\hat{\theta}_{MLE}$ 和矩估计量 $\hat{\theta}_{MoM}$。
(b) 最大似然估计量 $\hat{\theta}_{MLE}$ 是无偏的吗？
(c) 计算 θ 的 $C-R$ 下界。$C-R$ 下界能达到吗？

4. 设 $X_1,\ldots,X_n$ 是从参数为 θ 的指数分布中独立同分布抽取的样本（$f_\theta(x)=\theta\exp(-\theta x), x\geq 0$）。记 $F_\theta(t)$ 为该指数分布的累积分布函数，$t>0$。求尾概率 $p=1-F_\theta(t)$ 这一参数的 UMVUE。

5. 假设 $X_1,\ldots,X_n$ 是来自总体族 P_θ 的独立同分布样本，$T_n=T_n(X_1,\ldots,X_n)$ 是 $g(\theta)$ 的一个估计量。如果对于某些 $a_k(\theta)$，有

$$E_\theta[T_n]-g(\theta)=\sum_{k=1}^{\infty}\frac{a_k(\theta)}{n^k}$$

即 $E_\theta[T_n]-g(\theta)=O\big(\frac{1}{n}\big)$。令 $T_{n-1}^{(-i)}$ 表示 $T_{n-1}^{-1}(X_1,\dots,X_{i-1},X_{i+1},\dots,X_n)$，并定义

$$\bar{T}_{n-1}=\frac{1}{n}\sum_{i=1}^{n}T_{n-1}^{(-i)},$$

$$T_n'=nT_n-(n-1)\bar{T}_{n-1},$$

其中 T_n' 称为 T_n 的 Jackknife 估计量。

(a) 证明：Jackknife 估计量可以减小原始估计量 T_n 的偏差。

$$\mathbb{E}_\theta[T_n']-g(\theta)=O\left(\frac{1}{n^2}\right)$$

(b) 现在假设 $X_1,X_2,\dots,X_n$ 服从 $N(\theta,1)$，我们想估计 μ^2。一个自然的估计量是 $T_n=\bar{X}^2=\big(\frac{1}{n}\sum_{k=1}^{2}X_k\big)^2$。使用上述符号推导出 Jackknife 估计量 T_n'。并计算偏差 $\mathbb{E}[T_n]-\mu^2$、$\mathbb{E}[T_n']-\mu^2$ 和方差 $\mathrm{Var}(T_n),\mathrm{Var}(T_n')$. 比较结果以说明，虽然 Jackknife 估计量的偏差更小，但它可能会增大方差。

1.11　2025 年春季数理统计期末考试模拟试卷

1. 设 X_1,X_2 独立同分布，其共同密度函数为

$$p(x;\theta)=k\theta^k x^{-(k+1)},\quad x>\theta,\quad \theta>0,\quad k>2\text{ 已知.}$$

   (1) 证明 $T_1=\frac{k-1}{2k}(X_1+X_2)$ 和 $T_2=\frac{2k-1}{2k}\min(X_1,X_2)$ 都是 θ 的无偏估计；

   (2) 计算 T_1 和 T_2 的均方误差并进行比较；

   (3) 证明：在均方误差意义下，在形如 $T_c=c\min(X_1,X_2)$ 的估计中，$c=\frac{2k-2}{2k-1}$ 时最优。

2. 设 $X_1,\dots,X_n$ 为独立同分布随机变量，$0<\theta<1$，

$$P(X_1=-1)=\frac{1-\theta}{2},\quad P(X_1=0)=1/2,\quad P(X_1=1)=\frac{\theta}{2}$$

   (1) 求 θ 的 MLE $\hat{\theta}_1$ 并问 $\hat{\theta}_1$ 是否是无偏的；

   (2) 求 θ 的矩估计 $\hat{\theta}_2$；

   (3) 计算 θ 的无偏估计的方差的 C–R 下界。

3. 设 $X_1,\dots,X_n$ 是来自 $U(\theta_1,\theta_2)$ 的样本。

   (1) 证明 $(X_{(1)},X_{(n)})$ 是该分布族的完备充分统计量；

   (2) 求 θ_1,θ_2 的一致最小方差无偏估计；

   (3) 证明：若 $\theta_2=2\theta_1$，则 $(X_{(1)},X_{(n)})$ 不再是完备的；

   (4) 对 $\theta_2=2\theta_1$，求 θ_1 的 MLE，记为 $\hat{\theta}_2$；

   (5) 对 $\theta_2=2\theta_1$，令 $T=\frac{(n+1)(2X_{(n)}+X_{(1)})}{5n+4}$，则 T 是 θ_1 的无偏估计，且 $\mathrm{Var}(T)\le\mathrm{Var}(\hat{\theta}_1)$。

4. 设 $X_1,\dots,X_n$ 和 $Y_1,\dots,Y_m$ 分别为来自参数为 λ_1 和 λ_2 的指数分布的样本，且全样本独立，试求 λ_2/λ_1 的置信水平为 $1-\alpha$ 的置信区间。

5. 设 $X_1,\dots,X_n$ 是来自概率密度函数为

$$f(x;\theta)=2\frac{x}{\theta^2}\exp\left(-\frac{x^2}{\theta^2}\right),\quad x>0$$

   的 Rayleigh 分布的样本，其中 $\theta>0$ 为未知参数。

(1) 利用此总体的充分完备统计量，建立一个关于假设

$$H_0 : \theta = 1 \quad \text{vs} \quad H_1 : \theta > 1$$

的显著性水平为 α 的检验；

(2) 验证此检验统计量在 H_1 下比在 H_0 下具有更大的均值。

6. 假设 $X_{i1}, \ldots, X_{in_i}, i = 1, 2$ 为两个 IID 样本，分别服从均匀分布 $U(0, \theta_i), i = 1, 2$，这里 $\theta_1 > 0, \theta_2 > 0$ 均为未知。

(1) 求假设 $H_0 : \theta_1 = \theta_2$ vs $H_1 : \theta_1 \neq \theta_2$ 的显著性水平为 α 的似然比检验；

(2) 求 $-2 \ln \lambda$ 的极限分布，这里 λ 为 (1) 中的似然比。

7. 设 $X_1, \ldots, X_n$ 为来自 $b(1, p)$ 的样本，试利用 N–P 引理求假设

$$H_0 : p = p_0 \quad \text{vs} \quad H_1 : p = p_1 (> p_0)$$

的显著性水平为 α 的最大功效检验。

8. 设为一位顾客的服务时间服从指数分布 Exp(λ)，其中 λ 未知，又设 λ 的先验分布是均值为 0.2，标准差为 1 的 Gamma 分布。如今对 20 位顾客服务，平均服务时间是 3.8 分钟。在平方损失下，分别求 λ 和 $\theta = 1/\lambda$ 的 Bayes 估计。

(参数 λ 在平方损失下的 Bayes 估计满足 $\bar{\lambda} = \underset{\hat{\lambda}}{\text{argmin}} E[(\hat{\lambda} - \lambda)^2 | X]$.)

9. 某建筑工地每天发生事故数现场记录如下：

一天发生的事故数	0	1	2	3	4	5	≥ 6	合计
天数	102	59	30	8	0	1	0	200

试在显著性水平 $\alpha = 0.05$ 下检验这批数据是否服从泊松分布。

10. 为了比较用来做鞋后跟的两种材料的质量，选取了 15 个男子 (他们生活条件各不相同)，每人穿着一双新鞋，其中一只是材料 A 做后跟，另一只以材料 B 做后跟，其厚度均为 10mm，过了一个月再测量厚度，得到数据如下：

序号	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
材料 A	6.6	7.0	8.3	8.2	5.2	9.3	7.9	8.5	7.8	7.5	6.1	8.9	6.1	9.4	9.1
材料 B	7.4	5.4	8.8	8.0	6.8	9.1	6.3	7.5	7.0	6.5	4.4	7.7	4.2	9.4	9.1

问是否可以认定材料 A 制成的后跟比材料 B 耐穿？

(1) 设 $d_i = x_i - y_i, i = 1, 2, \ldots, 15$ 来自正态总体，结论是什么？

(2) 采用符号秩和检验方法检验，结论是什么？